

Claim Standing and Legitimacy

A Domain-General Audit Protocol for Laundering Exclusion and Reportable Output

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May 2026

Abstract

Claims often fail not because they are false, but because they are not yet well-formed objects of responsible assertion: their target drifts, their support carrier is unfixed, their evidence is proxy-contaminated, or their reported status exceeds what the audit supports. This paper develops the Universal Exactness Audit Protocol (UEAP) as a domain-general protocol for assigning the strongest responsible status of a claim.

The main result is status-relative. For a domain D , claim c , and requested status class σ , UEAP-legitimacy at σ is certified exactly when target fixation, carrier adequacy, exhaustion of admissible alternatives modulo skin, laundering exclusion, and report-preserving output all hold:

$$\begin{aligned} \text{UEAPLegit}_D^\sigma(c) \iff & \text{TargetFix}_D(c) \wedge \text{CarrierStand}_D^\sigma(c) \\ & \wedge \text{Exhaust}_D^\sigma(\text{AdmAlt}(c)/\sim_{\text{skin}}) \wedge \text{NoLaunder}_D(c) \\ & \wedge \text{Report}_D^\sigma(c). \end{aligned}$$

The paper proves this as a conditional protocol-level result from a pre-audit observation frame and a coordinate-exhaustion theorem: once the audit basis is declared, any status-relevant claim variation must appear in target, carrier, semantic content, representation, ancestry, domain, evidence network, report/use profile, or requested status class. The certification condition is internal to UEAP-legitimacy at the requested status; it is not a sociological, ethical, institutional, or metaphysical definition of legitimacy in general.

The manuscript gives the formal audit objects, validator, registry, status-precedence rule, laundering taxonomy, and domain instantiation guides. Three worked audits demonstrate the protocol: Black-Scholes valuation as representative-invariant local valuation, self-adjoint extension as endpoint selection versus auxiliary-data import, and cosmological transport closure as disciplined numerical reportability. UEAP is not a truth oracle and does not replace domain expertise; it classifies standing, blockers, and reportability so that claims are not promoted beyond what their carrier and ledger support.

Keywords. claim standing; legitimacy; admissibility; laundering exclusion; exactness; reportability; validation; evidence networks; calibration; audit protocol.

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1 Introduction: the claim-legitimacy problem

1.1 Motivation

Many claims fail before truth or falsity can be responsibly assessed. A string of words may not fix a target. A model output may not carry standing. A numerical match may be calibrated to the outcome it appears to predict. A proof-like derivation may use a later repair as if it had always been the same admissible act. A scientific comparison may treat two domain-relative estimates as the same parameter without a transport ledger. A legal or institutional claim may cite authority while leaving its evidential carrier unfixed.

These failures are not merely weak support. They are failures of claim standing. A claim that changes under admissible redescription, depends on hidden target ancestry, relies on an undeclared boundary choice, or reports a stronger status than its carrier supports has not yet reached the stage at which final legitimacy can be assigned.

The Universal Exactness Audit Protocol, abbreviated UEAP, is developed in the present paper from the exactness arc [13]; it is a domain-general protocol for such cases. It begins before exactness. Its first question is not whether a claim is exact, true, validated, proven, actionable, or useful. Its first question is whether the claim has a fixed target and a standing-bearing carrier. Only after target and carrier are fixed can the audit move to alternatives, laundering, ancestry, evidence networks, and reportability.

1.2 Core thesis

The governing principle, stated as the operational core of UEAP in the present paper, is:

$$\begin{aligned} \text{UEAPLegit}_D^\sigma(c) \iff & \text{TargetFix}_D(c) \wedge \text{CarrierStand}_D^\sigma(c) \\ & \wedge \text{Exhaust}_D^\sigma(\text{AdmAlt}(c)/\sim_{\text{Skin}}) \wedge \text{NoLaunder}_D(c) \\ & \wedge \text{Report}_D^\sigma(c). \end{aligned}$$

Here σ is the requested status class: legitimate-final, legitimate-but-not-actionable, relation-only, class-level, obstruction-only, calibrated-only, domain-dependent, evidence-compatible, computation-only, failed, or another declared domain status. The notation $\text{UEAPLegit}_D^\sigma(c)$ denotes UEAP-legitimacy at requested status σ , not every possible social, ethical, institutional, or pragmatic sense of legitimacy. The biconditional is therefore internal to UEAP-legitimacy: it is the certification condition once D , σ , carrier standards, admissible alternatives, skin relation, laundering taxonomy, and precedence order have been declared. This expression is stronger than a checklist and narrower than a truth oracle. It says that final assertion at a requested status is available exactly when the audit conditions required for that status are jointly satisfied.

Exact numerical prediction is one strong output class of this protocol. It is not the master object. A claim may be legitimate but not actionable, compatible only, calibrated only, domain-dependent, relation-only, class-level only, computation-only, obstruction-only, or failed. The protocol's purpose is to assign the strongest defensible status rather than to force all claims into a binary true/false or exact/inexact mold.

The status-relative form is essential. UEAP does not assert that the same evidence carrier suffices across medicine, law, mathematical proof, model validation, engineering safety, finance, or cosmological inference. Domain expertise determines the content of the carrier and the evidence standard for σ . UEAP determines whether the asserted status is licensed by the target, carrier, freedom ledger, laundering audit, and report object.

1.3 Contribution

The paper contributes five things.

First, it gives a claim-standing account of legitimacy. A claim must have a fixed target role and a standing-bearing carrier before it can enter a final legitimacy audit.

Second, it generalizes exactness ledgers into freedom ledgers. A claim remains unfixed when admissible variations can change target, carrier, meaning, scope, representation, context, baseline, calibration ancestry, evidence independence, or reportability.

Third, it formalizes laundering as a central audit blocker. A transformation cannot promote claim status unless it explicitly discharges the freedom it appears to remove. Otherwise the transformed claim inherits the original blocker.

Fourth, it gives an operational status taxonomy, validator protocol, and registry schema. The result is not only a philosophical classification but a repeatable audit practice.

Fifth, it demonstrates the method on three domains: mathematical finance, operator-theoretic mathematical physics, and empirical cosmological inference. The pilots are intentionally different. Their common structure shows that UEAP is a claim-standing protocol rather than a topic-specific checklist.

1.4 Nonclaims

Nonclaim 1.1 (Limits of UEAP). UEAP is not a truth oracle, not a replacement for domain expertise, not a model-selection rule, not a proof assistant, not a statistical significance test, and not a way to make unsupported claims legitimate by formal vocabulary. It classifies standing, support, freedom, laundering, and reportability status.

Some true claims may remain under-legitimated because their carrier is not fixed. Some useful claims may remain calibrated-only or compatible-only. Some exact obstructions may support no positive assertion. Some domain-dependent claims may be fully legitimate only after their domain is declared as input rather than treated as a conclusion.

1.5 Typical claim failures

UEAP is motivated by a recurring pattern. A claim can be well formed but not standing-fixed; supported but not target-aligned; numerically impressive but calibrated-only; evidence-rich but not independent; domain-accurate but over-reported as universal; or computationally precise while carrier-unfixed. These are different failures, and treating them all as ordinary uncertainty loses information. UEAP therefore separates four questions that are often collapsed:

1. What is the target of the claim?
2. What carries the claim's standing?
3. What unresolved freedoms could change the claim?
4. What status can be reported without promotion?

The audit does not assume that every claim should be upgraded to final legitimacy. It is equally important to preserve lower statuses. A calibrated-only claim can be scientifically useful; a domain-dependent claim can be operationally valuable; an obstruction-only claim can be decisive against a

route while not providing a positive replacement; an evidence-compatible claim can guide future work without pretending to close alternatives.

1.6 A note on terminology

The word “legitimate” is used in a technical audit sense. It does not mean morally approved, institutionally accepted, socially credible, or true. It means that the claim has the standing, closure, no-laundering, and reportability conditions appropriate to the status being asserted. A false claim may be legitimately asserted as a failed claim if its failure is exactly certified. A true claim may remain illegitimate as a final assertion if its carrier or target has not been fixed.

2 From exactness to claim standing

2.1 The exactness arc as a special case

The original exactness arc asked when a numerical object becomes a reportable prediction [13]. UEAP extracts the broader protocol. The general problem is not confined to numerical values. A claim can be a value, relation, classification, explanation, causal assertion, proof-status report, model comparison, safety margin, legal conclusion, benchmark claim, or physical-parameter identification.

The translation is displayed in Table 1. Exactness-language objects become claim-audit objects once the target is generalized.

Exactness-arc object	UEAP object	General meaning
Numerical candidate	Claim	The assertion under audit.
Raw exactness	Legitimate final assertion	The claim is fixed in the requested strength class.
Quotient/representative freedom	Framing or representation ambiguity	The claim changes under redescription, encoding, wording, or presentation.
Transport freedom	Context/regime-transfer ambiguity	The claim changes across domains, scales, schemes, or contexts.
Scale/normalization freedom	Baseline, metric, unit, or score ambiguity	The reported magnitude depends on a convention.
Calibration ancestry	Target-derived or fitted support	The answer is inherited from the outcome or proxy.
Endpoint/domain freedom	Boundary or scope dependence	The claim depends on supplied domain, endpoint, or initial-condition choices.
Residue/finite invariant	Class-level or nonlocal output	A class, obstruction, residue, or relation is fixed rather than a local assertion.
Cross-sector exactness	Evidence network closure	Several constraints jointly eliminate alternatives without amplification.
Extraction	Reportability/actionability	The fixed claim can be stated or used without changing status.

Table 1: Translation from the exactness arc to the general UEAP claim audit.

2.2 Kernel-to-protocol bridge

UEAP is the operational form of a deeper admissible-construction kernel [14]. The kernel roles are reference, standing, admissibility, and irreversibility. UEAP translates these into claim-audit roles as shown in Table 2.

Kernel role	UEAP audit role	Failure output
Reference	Target fixation: the claim must preserve a determinate object, relation, value, event, or status under admissible redescription	target-unfixed, meaning-unfixed, scope-unfixed
Standing	Carrier standing: the support structure must be eligible to bear claim-status	carrier-unfixed, computation-only, authority-only
Admissibility	Alternative exhaustion and licensed continuation: unresolved alternatives cannot remain capable of changing the claim	compatible-only, evidence-compatible, failed
Irreversibility	No laundering and inherited blockers: later transformation cannot repair an unfixed claim as the same claim without discharging the blocker	laundered, calibrated-only, blocked

Table 2: Operational bridge from the admissible-construction kernel to UEAP.

The paper does not re-prove the entire non-derivability and minimality theory of the kernel. It uses the kernel as the minimal rationale for why target, carrier, admissibility, and no-repair conditions must be visible in a claim audit. The operational protocol remains self-contained: even a reader who brackets the foundational proof can apply the validator, registry, status taxonomy, and pilot audits.

2.3 Why the bridge matters

Without the kernel bridge, UEAP risks looking like a checklist. With the bridge, the checklist is presented as downstream from four claim roles that the kernel treats as necessary. Target fixation is the claim-level form of reference. Carrier standing is the claim-level form of eligible act-status. Freedom exhaustion is the claim-level form of admissibility. Laundering exclusion is the claim-level form of irreversibility.

Thus UEAP does not add a policy layer on top of claims. It operationalizes the minimal conditions under which a claim can bear legitimacy at all.

2.4 Operational independence from the foundational proof

The foundational kernel can be studied as a separate necessity theorem. UEAP does not require the reader to accept that theorem in order to use the protocol. The operational claims are weaker and directly inspectable: a target-unfixed claim cannot be audited as a fixed claim; a carrier-unfixed output cannot be reported as standing-bearing support; an unresolved admissible alternative blocks final assertion; and a transformation that hides rather than discharges a blocker cannot increase status. The kernel explains why these requirements are not optional. The validator shows how to apply them.

This separation fixes the scope of the present contribution. The present paper establishes the operational audit protocol and its cross-domain demonstrations while treating the kernel as a stated foundational rationale rather than rederiving it in full. The kernel remains visible in the protocol’s architecture, but the immediate contribution is the validator, registry, status taxonomy, and worked audit practice.

2.5 Proof-status architecture: definitions, protocol theorems, kernel theorems

A protocol paper of this scope must not hide the proof status of its claims. UEAP therefore uses three layers.

Layer	Role	Examples in this manuscript
Definition	Fixes an audit object or status rule.	Claim, target, carrier, freedom ledger, skin, report object, blocker set.
Protocol theorem	Derives a status consequence from the UEAP audit objects.	Freedom exhaustion, laundering exclusion, calibration-only, boundary declaration, report preservation, unique status assignment.
Kernel theorem	Supplies the deeper rationale for why the audit roles are not optional.	Reference as target fixation, standing as carrier eligibility, admissibility as licensed continuation, irreversibility as no same-act repair.

The manuscript proves its protocol theorems explicitly from the audit objects it defines. It does not pretend that every protocol theorem is a foundational theorem of the kernel. Conversely, it does not reduce the protocol to vocabulary choice: once the audit objects are fixed, the theorem proofs constrain how a claim can move through the status taxonomy. These are conditional protocol theorems. They do not assert that no alternative social or institutional use of the word legitimacy can be introduced outside UEAP; they assert that, under a declared UEAP audit basis, no status transition is licensed outside the registered coordinates. A reader may reject a protocol theorem only by changing the audit basis, denying one of the audit objects, denying the status-precedence rule, or denying the kernel bridge that motivates the object.

2.6 Requested status class and domain standards

Let σ denote the status being asserted. A claim may be legitimate as relation-only while not legitimate as a numerical prediction; legitimate as domain-dependent while not legitimate as universal; legitimate as obstruction-only while not legitimate as a positive construction. Carrier adequacy and alternative exhaustion are therefore indexed to σ . This prevents two opposite errors: over-universalizing the evidence standard, and allowing a weaker carrier to be laundered into a stronger report.

2.7 Strength classes

UEAP does not collapse all statuses into one binary. The requested strength of a claim determines what must be closed. A relation-only claim does not need a member selector; a class-level claim does not need a local coordinate; an obstruction-only claim does not need a positive construction; a calibrated-only claim does not need predictive ancestry. The problem arises only when a claim is reported at a strength

whose preconditions are not satisfied. The status taxonomy in Table 6 is therefore status-preserving rather than status-maximizing.

3 Core audit objects

Definition 3.1 (Claim). A claim c in a domain D is an assertion offered for standing-bearing evaluation. The claim may be verbal, mathematical, numerical, computational, empirical, legal, institutional, or operational. It is not required to be true, false, exact, or action-ready at the moment it enters the audit.

Definition 3.2 (Target role). The target role of a claim is

$$\text{Target}_D(c),$$

the object, relation, event, status, class, value, property, parameter, or inference that the claim purports to assert. A target is fixed when admissible redescription preserves what is being asserted.

Definition 3.3 (Carrier). The carrier of a claim is

$$\text{Carrier}_D(c),$$

the structure that purports to give the claim standing. Depending on the domain, the carrier may be a proof, derivation, measurement protocol, model interface, data ledger, estimator, quotient, operator, transport map, evidence chain, boundary record, registry entry, or experimental design.

Criterion 3.4 (Carrier Adequacy Test). A carrier $\text{Carrier}_D(c)$ is standing-bearing for claim c at status class σ only if it satisfies all six tests below.

1. **Target alignment.** The carrier supports the same target asserted by c , not a proxy target or nearby conclusion.
2. **Role adequacy.** The carrier is the right kind of support for σ : proof for proof-status, measurement ledger for measurement-status, transport map for transport-status, evidence chain for forensic status, and so on.
3. **Status-class adequacy.** The carrier must discharge the blocker that separates the requested status class from the next weaker relevant status. A correlation may be adequate for evidence-compatible status but not for causal status; a derivation trace may be adequate for syntactic derivability but not for closure-stable theorem status unless the uptake and stability conditions are met.
4. **Redescription stability.** The support relation survives admissible rewording, recoding, coordinate change, or presentation change.
5. **No hidden selector.** The support does not depend on undeclared auxiliary choices, boundary conventions, representative choices, or evaluator rules.
6. **No same-act repair.** Later correction does not make the original unsupported carrier standing-bearing as the same carrier; it creates a fresh carrier or a fresh audit state.

Failure of any clause yields carrier-unfixed, domain-dependent, calibrated-only, computation-only, externally-selected, or failed status according to the blocker that caused failure.

Definition 3.5 (Freedom ledger). The freedom ledger of a claim is

$$\mu_D(c) = \{\mu_{\text{target}}, \mu_{\text{carrier}}, \mu_{\text{meaning}}, \mu_{\text{scope}}, \mu_{\text{rep}}, \mu_{\text{context}}, \mu_{\text{baseline}}, \mu, \mu_{\text{evidence}}, \mu_{\text{report}}\}.$$

A component is active when admissible variation of that type can change the claim’s content, support, scope, status, or reportable form.

Definition 3.6 (Skin relation). Two presentations of a claim are skin-equivalent, written $c_1 \sim_{\text{Skin}} c_2$, when their difference changes only wording, notation, formatting, harmless encoding, chart choice, line labels, or representative convention while preserving target, carrier, scope, ancestry, evidence status, and reportable output.

Definition 3.7 (Admissible realization set). The admissible realization set $\text{AdmReal}_D(c)$ is the class of claim-realizations that remain after target, carrier, domain, ancestry, alternative, laundering, evidence, and reportability constraints have been applied.

Definition 3.8 (Claim fixation). A claim is fixed at the requested strength when

$$|\text{AdmReal}_D(c)/\sim_{\text{Skin}}| = 1.$$

If this condition fails, the claim may still have a weaker status: compatible-only, calibrated-only, relation-only, class-level only, domain-dependent, evidence-compatible, computation-only, obstruction-only, or failed.

Definition 3.9 (Report object). The report object $\text{Report}_D(c)$ is the strongest statement that can be responsibly made about c without changing its audited status. It records final status, domain, dependencies, blockers, uncertainty, and nonclaim boundaries.

Definition 3.10 (Declared input). A datum is a declared input when it is explicitly included in the claim’s domain ledger before status assignment. Declared inputs can support conditional or domain-dependent claims, but they cannot be reported as conclusions of the claim they condition.

Definition 3.11 (Blocker). A blocker is an unresolved audit condition that prevents promotion to a stronger status. Blockers are not all failures: an active calibration blocker may yield calibrated-only status, an active domain blocker may yield domain-dependent status, and an active evidence-network blocker may yield evidence-compatible status.

Definition 3.12 (Actionability). A claim is actionable when its report object licenses a decision, intervention, policy, engineering use, legal use, model update, or downstream inference without changing its status. Actionability is stronger than fixation. A claim may be fixed but not actionable if its domain, uncertainty, risk, or nonclaim boundary prevents responsible operational use.

Remark 3.13 (Claim text versus claim content). The literal text of a claim is not the same as its audited content. Content is fixed by target, carrier, scope, ancestry, alternatives, and reportability. Two different texts may be the same claim under skin-equivalence; the same text may express different claims under target or scope drift.

4 Pre-audit observation frame and coordinate exhaustion

4.1 Raw frame before status vocabulary

UEAP is not allowed to win by naming its preferred audit categories and then declaring all rivals irrelevant. The protocol therefore begins with a pre-audit observation frame: the information a critic must be able to hold fixed when claiming that a final report status has changed.

Definition 4.1 (Pre-audit claim observation frame). For a candidate claim c , a pre-audit observation frame is a tuple

$$\mathcal{F}_c = (T, C, S, R, A, D, E, P, U),$$

where T records the target object or target role; C records the carrier or support candidate; S records semantic content and meaning relations; R records representation, notation, redescription, and skin candidates; A records ancestry, calibration, tuning, proxy, and outcome-conditioned selection structure; D records domain, scope, boundary, population, regime, endpoint, metric, and baseline data; E records evidence nodes and dependence relations; P records report, action, visualization, decision, and downstream-use profile; and U records the requested uptake or status class.

The frame is descriptive rather than justificatory. It records what the claim is about, what purports to support it, how it is presented, where it applies, where its support came from, what evidence is connected to it, and how it is to be used or reported. It does not yet declare the claim legitimate.

Definition 4.2 (Status-relevant variation). A variation v of a pre-audit frame is status-relevant when replacing $\mathcal{F}_c$ by $v(\mathcal{F}_c)$ can change at least one of the following: the strongest defensible report status, the blocker set, the next-action requirement, the admissible realization class, or the downstream use profile licensed by the report.

Definition 4.3 (Pre-audit skin). A variation is pre-audit skin when it changes only labels, typography, harmless notation, citation numbering, display order, or presentation format while preserving every coordinate of $\mathcal{F}_c$ up to the equivalences declared in the frame. Pre-audit skin is therefore stronger than visual similarity and weaker than content identity: it preserves all status-relevant coordinates but not necessarily all surface features.

4.2 Coordinate exhaustion

The coordinates in $\mathcal{F}_c$ are not an arbitrary catalogue. They are the finite loci through which a claim-status change can enter. A status change must alter what is asserted, what supports it, what it means, how representation is identified, where the support came from, where the claim applies, what evidence network is being used, how it is reported or used, or which status class is being requested.

Theorem 4.4 (Claim-Failure Coordinate Exhaustion). *Let v be any non-skin variation of a pre-audit frame $\mathcal{F}_c$. If v can change the strongest defensible report status of c , then v alters at least one coordinate of*

$$(T, C, S, R, A, D, E, P, U).$$

Equivalently, any status-relevant claim failure must appear as target failure, carrier failure, semantic failure, representative/redescription failure, ancestry/calibration failure, domain/scope/boundary failure, evidence-network failure, report/use failure, or requested-status failure.

Proof. A report status is a status of a claim under a requested use. A status-relevant variation must therefore change some determinant of the claim or its permitted report. If it changes what is asserted, it alters T . If it changes what bears support, it alters C . If the same words now express different content, it alters S . If the claimed sameness of presentations changes, it alters R . If the answer or support comes from a different target, proxy, calibration, tuning, or outcome-conditioned path, it alters A . If the claim applies under a different domain, endpoint, boundary, population, regime, metric, or baseline, it alters D . If the supporting sources, dependencies, independence relations, or compatibility structure change, it alters E . If the statement, action, visualization, or downstream use changes what is licensed, it alters P . If the requested strength changes, it alters U . These cases exhaust the ways

a status can change: the status concerns an asserted target, a carrier, a meaning, a representation class, an ancestry path, a domain, an evidence network, a report/use profile, and a requested uptake class. If none of these coordinates changes, then the same target, carrier, meaning, representation class, ancestry, domain, evidence network, report/use, and requested status are being evaluated. The variation is pre-audit skin and cannot change the strongest defensible report status. Therefore every status-relevant non-skin variation alters at least one coordinate. $\square$

Corollary 4.5 (No missing primitive coordinate). *A proposed additional audit coordinate is either a specialization of one of $T, C, S, R, A, D, E, P, U$, a declared refinement of a domain-specific carrier standard, or status-inert. For example, risk and cost belong to the report/use profile or the requested actionability standard; ethical admissibility belongs to the domain ledger or requested status class; replication status belongs to the evidence network or carrier standard; causal mechanism belongs to target, carrier, or evidence structure; robustness class belongs to the domain ledger, evidence network, or report object.*

Proof. If the proposed coordinate can change status, it must change what is asserted, what supports it, how it is interpreted or represented, where it came from, where it applies, what evidence structure it has, how it is reported or used, or which status is requested. By the theorem it is therefore a refinement of one of the listed coordinates. If it changes none of those, it cannot change status and is status-inert. $\square$

Principle 4.6 (Counterexample burden). *A counterexample to UEAP must exhibit a claim whose strongest defensible report status changes while target, carrier, semantic content, representation class, ancestry, domain, evidence network, report/use profile, requested status class, admissible alternatives, laundering status, and blocker set are all held fixed. A useful process that changes one of those coordinates is not a counterexample; it identifies the coordinate through which the status changed.*

Remark 4.7 (Why this is the central spine). The observation-frame theorem is the bridge from protocol usefulness to protocol completeness. Without it, UEAP could be read as a disciplined checklist. With it, the checklist becomes the operational interface of a coordinate-exhaustion result: all status-relevant failures must enter through the frame coordinates and therefore can be registered, blocked, discharged, or reported at a weaker status.

5 Claim standing and target fixation

5.1 Standing before exactness

A claim must be capable of bearing legitimacy before exactness, validation, proof, or failure can be assigned. A string, model output, institutional statement, graph, data fit, or numerical value is not automatically a claim with standing. It becomes audit-ready only when the target role and carrier role are fixed enough to identify what is being evaluated and what structure bears the evaluation.

Theorem 5.1 (Claim Standing Necessity). *A claim c cannot be evaluated as legitimate, exact, calibrated, failed, obstruction-only, domain-dependent, or reportable on D until both $\text{Target}_D(c)$ and $\text{Carrier}_D(c)$ are fixed up to skin-equivalence. If target or carrier drift remains active, the claim is standing-unfixed, not merely weakly supported.*

Proof. If $\text{Target}_D(c)$ is unfixed, there is no stable object of evaluation. A later verdict could apply to a different value, event, relation, parameter, scope, or meaning while using the same wording. That would

evaluate a moving target rather than a claim. If $\text{Carrier}_D(c)$ is unfixed, there is no standing-bearing structure capable of supporting the claim. A numerical match, authority citation, syntactic trace, or model output may be recorded, but it does not yet determine a licit claim-bearing support. The Carrier Adequacy Test identifies the missing structure: target alignment, role adequacy, status-class adequacy, redescription stability, absence of hidden selector, and no same-act repair. Exactness, calibration, failure, obstruction, and reportability all presuppose a fixed object and an eligible carrier. Therefore target and carrier fixation are necessary before the audit can assign any final or intermediate status. $\square$

Corollary 5.2 (Standing-unfixed is not weak support). *A target-unfixed or carrier-unfixed claim should not be reported as weakly legitimate. Its proper status is standing-unfixed until the target or carrier role is fixed.*

Definition 5.3 (Standing-status outputs). The preliminary standing statuses are:

Status	Meaning
standing-fixed	Target and carrier are fixed enough to audit the claim.
target-unfixed	The object of the claim can drift under admissible redescription.
carrier-unfixed	No standing-bearing support structure has been fixed.
scope-unfixed	Domain, boundary, population, or operating regime is not declared.
meaning-unfixed	Semantic content changes under admissible redescription.
audit-ready	The claim may proceed to freedom-ledger and laundering audit.

5.2 Examples of preliminary standing failures

A medical biomarker claim may be target-unfixed if the asserted target alternates between diagnosis, prognosis, treatment response, and surrogate endpoint. An AI benchmark claim may be carrier-unfixed if the benchmark score is reported without training-data ancestry, task definition, leakage audit, or evaluation protocol. A forensic reconstruction may be scope-unfixed if it silently changes from scene-level compatibility to individual causation. A cosmological parameter claim may be meaning-unfixed if early- and late-sector quantities are treated as the same parameter without a transport ledger.

These examples are not final audits. They show why UEAP begins before exactness. A claim whose object or carrier drifts cannot be rescued by adding more data to the same unstable presentation. The first audit task is stabilization, not accumulation.

Proposition 5.4 (Target-carrier alignment). *If a carrier supports a target different from the one asserted by c , then c is target/carrier-misaligned and cannot receive final legitimate status. The audit must either retarget the claim, replace the carrier, or report a weaker status.*

Proof. A carrier gives standing only to the target it supports. If the asserted target differs, the claim's reportable content is not licensed by its support. Treating the carrier as sufficient would silently shift either the target or the support role. That is target laundering or carrier laundering. Therefore final legitimacy is blocked until alignment is restored or the claim is reclassified. $\square$

6 Freedom ledgers and skin/tensor separation

6.1 Why freedom matters

A claim can remain superficially stable while its content changes. The same sentence may shift target, domain, calibration basis, evidence interpretation, or reportability. The freedom ledger records exactly those degrees of freedom that can change the claim while leaving its external presentation intact.

Definition 6.1 (Active freedom). A freedom component $\mu_i(c)$ is active when some admissible variation of type i changes the claim’s target, carrier, support, status, or reportable form. It is closed when every such variation is skin, inadmissible, declared input, obstruction-only, or uniquely fixed.

Table 3: Freedom ledger components.

Freedom	Audit question
μ_{target}	Can the object of the claim shift?
μ_{carrier}	Can the standing-bearing support structure change?
μ_{meaning}	Can semantic content change while wording remains stable?
μ_{scope}	Can domain, boundary, population, or operating regime change?
μ_{rep}	Can representation, notation, encoding, or framing change claim content?
μ_{context}	Does transfer across context, model regime, scale, or discipline alter the claim?
μ_{baseline}	Does a metric, norm, baseline, score, unit, or normalization determine the reported result?
μ	Was the result fitted, trained, tuned, or selected from target or proxy data?
μ_{evidence}	Do evidence sources share ancestry, target contamination, estimator dependence, or proxy leakage?
μ_{report}	Does reporting, actionability, visualization, or decision use alter the claim status?

Definition 6.2 (Skin and tensor variation). A variation is *skin* when it changes only presentation while preserving standing-bearing content. A variation is *tensor* when it changes content-bearing structure: target, carrier, scope, ancestry, evidence, quotient identity, boundary, estimator, or reportability.

Theorem 6.3 (Freedom Exhaustion Criterion). *A claim c is fixed at a requested strength only when every admissible alternative capable of changing c is classified as exactly one of the following: skin, inadmissible, obstruction-only, declared input, or uniquely fixed. If any content-changing admissible alternative remains unclassified, the claim cannot receive final legitimate status at that strength.*

Proof. Let a be an admissible alternative capable of changing the claim. If a is skin, it does not change standing-bearing content and can be quotient-collapsed. If a is inadmissible, it cannot compete within the domain. If a is obstruction-only, it supplies a negative result but no positive alternative assertion. If a is declared input, it becomes part of the domain ledger rather than a hidden freedom. If a is uniquely fixed, no rival alternative remains. These cases exhaust the ways an alternative can fail to

remain an open content-changing competitor. If an admissible content-changing alternative is not in one of these classes, then more than one non-skin realization remains in $\text{AdmReal}_D(c)/\sim_{\text{Skin}}$. Final legitimacy at the requested strength is therefore blocked. $\square$

Proposition 6.4 (Skin cannot increase legitimacy). *A skin transformation cannot increase a claim’s legitimacy status.*

Proof. By definition, skin preserves target, carrier, scope, ancestry, evidence status, and reportable form. Any increase in legitimacy would require a change in at least one status-bearing coordinate. Since skin changes none, it can only redescribe the same status. $\square$

Proposition 6.5 (Tensor variation must be declared or closed). *A tensor variation that changes standing-bearing content must be declared as an input, exhausted, uniquely fixed, or treated as a blocker. It cannot remain hidden while a stronger status is reported.*

Proof. A tensor variation changes content. If hidden, it leaves multiple non-skin realizations available while the claim is reported as if only one realization remained. That violates the freedom exhaustion criterion. Thus it must be declared, closed, or reported as a blocker. $\square$

6.2 Freedom normal forms

The freedom ledger produces normal forms for failure. Target freedom produces target drift; carrier freedom produces support drift; meaning freedom produces semantic laundering; scope freedom produces domain dependence; representation freedom produces representative dependence; context freedom produces transport failure; baseline freedom produces metric/baseline laundering; calibration freedom produces calibrated-only status; evidence freedom produces evidence-compatible status; report freedom produces over-reporting. The normal forms are useful because they tell the author what must be repaired or reclassified.

Active freedom	Normal form	Typical strongest status
Target	target drift	target-unfixed
Carrier	support drift	carrier-unfixed
Meaning	semantic redescription	meaning-unfixed
Scope	domain/scope drift	domain-dependent
Representation	representative dependence	compatible-only or failed
Context	transport failure	domain-dependent or failed
Baseline	metric or normalization dependence	domain-dependent
Calibration	target/proxy ancestry	calibrated-only
Evidence	non-independent support	evidence-compatible
Report	status-changing output	not reportable

Table 4: Freedom-ledger normal forms.

7 Laundering taxonomy and exclusion

7.1 Definition and purpose

Laundering is the central UEAP failure mode. It occurs when an unsupported or unfixed structure is passed through a transformation that makes it appear legitimate without discharging the relevant freedom. The transformation may be semantic, computational, statistical, institutional, evidential, or representational.

Definition 7.1 (Laundering). A claim c is laundered when some transformation L changes the apparent status of c while leaving an active freedom or blocker unresolved. In that case $L(c)$ inherits the original blocking status unless L explicitly discharges the freedom it purports to remove.

Table 5: UEAP laundering taxonomy.

Route	Failure mode	Required discharge
Semantic laundering	Meaning changes while wording or notation is preserved.	Fix semantic target.
Proxy laundering	A variable recovers the target while appearing independent.	Prove proxy independence or declare target ancestry.
Calibration laundering	Fitted or target-derived output is reported as prediction.	Separate calibration from prediction.
Domain laundering	Scope, population, boundary, or regime changes after support is known.	Declare and stabilize domain.
Endpoint laundering	Boundary or initial conditions are selected to recover the desired result.	Independently fix endpoints.
Evidence-network laundering	Non-independent supports are counted as global proof.	Audit ancestry and independence.
Computation laundering	Solver, simulation, decimal output, or benchmark is treated as legitimacy.	Establish target, carrier, and alternatives.
Authority laundering	Consensus, publication, expert confidence, or institutional status replaces standing.	Provide standing-bearing carrier.
Post-hoc explanation laundering	A target is explained after being observed and reported as prediction.	Show target-independent ancestry.
Metric/baseline laundering	A score, norm, baseline, or unit manufactures support.	Fix metric or report domain dependence.
Target-drift laundering	The claim target moves after evidence is known.	Restore target fixation.

7.2 Why laundering is stronger than error

An error is a wrong claim. A laundering route is a wrong status transition. A laundered claim may be based on a true observation, a valid computation, a useful fit, or a respected source. The problem is that the transformation promotes the claim to a status whose conditions were not discharged. This distinction explains why UEAP does not simply label all problematic claims false. It preserves lower statuses and blocks only the illegitimate promotion.

7.3 Discharging a laundering route

To discharge a laundering route, the author must do more than rename it. Semantic laundering is discharged by fixing meaning under redescription. Proxy laundering is discharged by proving proxy independence or declaring target ancestry. Calibration laundering is discharged by separating calibration from prediction. Domain laundering is discharged by fixing or declaring the domain. Computation laundering is discharged by supplying target and carrier, not by adding digits. Authority laundering is discharged by identifying the standing-bearing carrier behind the authority claim.

Theorem 7.2 (Laundering Exclusion). *A transformation cannot increase claim legitimacy unless it explicitly discharges the freedom it appears to remove. Otherwise the transformed claim inherits the original blocking status.*

Proof. Let L be a transformation applied to a claim c . If L discharges an active freedom, then it changes the audit state by moving a content-changing alternative into skin, inadmissible, declared input, obstruction-only, or uniquely fixed status. Legitimacy may then increase if all other conditions are met. If L does not discharge the freedom, the set $\text{AdmReal}_D(c)/\sim_{\text{Skin}}$ still contains the same unresolved content-changing alternative, or the same carrier/target/scope/ancestry blocker remains. Reporting a higher status for $L(c)$ would therefore treat an unresolved freedom as resolved. That is laundering. The proper status is inherited from the original blocker. $\square$

Corollary 7.3 (Computation alone is not legitimacy). *A numerical solver, simulation, proof-check trace, machine-learning output, benchmark score, graph, or decimal value does not confer standing merely by being computational. Computation may be part of a standing-bearing carrier when its target, domain, input ledger, admissible transformations, verification conditions, alternatives, ancestry, and report status are fixed. Without that carrier structure, the output remains computation-only.*

Corollary 7.4 (Authority alone is not standing). *Publication, consensus, expert confidence, institutional status, or legal authority does not by itself replace carrier standing. Authority can be evidence about a carrier, and in domains whose target is institutional authority it may itself be part of the target. It cannot substitute for the domain's required standing carrier when the target is mathematical, factual, causal, predictive, empirical, or operational.*

Corollary 7.5 (Calibration is not prediction). *A target-derived result remains calibrated-only unless the audit proves that the reported claim survives isolation, deletion, or independent replacement of target and proxy ancestry.*

8 Calibration, proxy, and ancestry graphs

8.1 Ancestry as a status coordinate

A claim can appear independently supported while inheriting its answer from the target. UEAP handles this through ancestry graphs.

Definition 8.1 (Ancestry graph). The ancestry graph of a claim is a directed graph

$$K_c = (V, E),$$

whose nodes include target data, proxy variables, fitted parameters, model assumptions, estimator choices, boundary inputs, evidence sources, and the final reported claim. Edges record dependency, fitting, conditioning, selection, calibration, or derivation.

Definition 8.2 (Target ancestry). A claim has target ancestry when a directed path from the target, a target proxy, outcome-conditioned selection, or target-derived calibration node reaches the final claim node.

Theorem 8.3 (Calibration-Only Theorem). *If a claim inherits its answer from target data or proxy data, then its strongest status is calibrated-only unless the claim survives removal, isolation, or independent replacement of the target/proxy nodes in the ancestry graph.*

Proof. Target or proxy ancestry means the result depends on information from the object it claims to predict, explain independently, classify, or infer. Unless that dependency is removed, isolated as a declared calibration input, or replaced by independent support, the result has not shown target-independent standing. Reporting it as prediction, discovery, independent validation, or proof would be calibration laundering. Therefore calibrated-only is the strongest status until the ancestry blocker is discharged. $\square$

8.2 Ancestry operations

UEAP uses three ancestry operations. The deletion test removes target and proxy nodes and asks whether the support path to the claim survives. The isolation test leaves target/proxy nodes in place but reports the result explicitly as calibrated. The replacement test substitutes independent target-free support and checks whether the claim remains fixed. These operations distinguish calibrated usefulness from predictive standing.

Proposition 8.4 (Deletion, isolation, replacement). *Let K_c be the ancestry graph of a claim. If every support path to the final claim node passes through target or proxy nodes, then deletion blocks predictive status. If those nodes are declared and isolated, calibrated-only status may be legitimate. If independent replacement paths fix the same claim, stronger status may be available.*

Proof. If deletion removes all support paths, the claim has no target-independent carrier. If the target/proxy nodes are declared, the audit can preserve the result as calibration rather than prediction. If independent replacement paths support the same claim, the target ancestry no longer carries the status. These are the three possible dispositions of target ancestry. $\square$

Corollary 8.5 (Retrospective explanation is not prediction). *Explaining a target after observing it does not support a predictive status unless the ancestry graph shows that the explanation's support path is independent of the observed target or that the target contribution is declared as calibration.*

Proposition 8.6 (Proxy leakage). *If a proxy variable reconstructs the target under an apparently independent name, then the claim inherits target ancestry unless the proxy’s independence is established in the ancestry graph.*

9 Domain, scope, and boundary closure

9.1 Domain ledgers

Claims are rarely domain-free. They may depend on population, scale, boundary condition, measurement regime, convention, numerical tolerance, operating environment, horizon, calibration standard, legal jurisdiction, or physical interface. UEAP does not forbid domain dependence. It requires domain dependence to be declared and preserved.

Definition 9.1 (Domain ledger). The domain ledger $\text{Dom}_D(c)$ records the boundary conditions, scope, population, regime, metric, endpoint, standard, normalization, tolerance envelope, and operating assumptions required for the claim.

Theorem 9.2 (Boundary Declaration Principle). *Scope, boundary, endpoint, and operating-regime choices cannot support a claim unless they are declared, stable, and standing-bearing. Otherwise they are inputs rather than conclusions, and the strongest status of the claim is domain-dependent or blocked.*

Proof. A boundary or endpoint choice that affects the claim is a content-bearing input. If it is not declared, the claim’s truth or support can change under a hidden domain variation. If it is declared but not standing-bearing, it may define a hypothetical or conditional analysis but not a final assertion about the target. If it is declared, stable, and standing-bearing, it can be part of the carrier or domain ledger. Thus undeclared or unsupported boundary choices cannot promote the claim; they must be reported as inputs, domain dependence, or blockers. $\square$

9.2 Inputs and conclusions

A domain choice may be legitimate as an input while illegitimate as a conclusion. For example, a safety-margin claim may be fully legitimate within a declared load envelope, but it cannot report that the envelope itself was proven by the test unless that was separately audited. A fitted cosmological transport basis may condition a constrained posterior, but the basis cannot be reported as discovered if it was supplied as a modeling input. A boundary condition in operator theory may select an extension, but the selection is not internal to the old operator base unless the boundary condition has standing there.

Corollary 9.3 (Endpoint laundering). *Selecting boundary or initial conditions to recover a desired result is not legitimate support unless those conditions are independently fixed, declared as inputs, or classified as a separate target.*

Corollary 9.4 (Domain-dependent status). *If a claim is fixed only under a supplied boundary, population, scope, or operating regime, its strongest general status is domain-dependent unless that boundary is itself justified as part of the claim.*

10 Evidence networks and no illicit status-amplification

10.1 Evidence is not additive by appearance

Multiple evidence sources can agree because they are independent confirmations, or because they share target ancestry, estimator dependence, calibration leakage, baseline choice, institutional authority, or common proxy structure. UEAP treats evidence as a network rather than a count. The point is not that evidence cannot accumulate. The point is that accumulation has to be represented as network closure rather than as a count of agreeing appearances.

Definition 10.1 (Evidence network). The evidence network $\text{Net}_D(c)$ is a graph of sources, dependencies, overlap relations, shared ancestry, estimator links, boundary assumptions, proxy contamination, and compatibility constraints relevant to the claim.

Definition 10.2 (Independent support). Support nodes are independent relative to c only if they do not share target ancestry, proxy contamination, estimator dependence, boundary selection, hidden common assumptions, or reportability-transforming baselines that affect the claim.

10.2 Evidence-network failure modes

Evidence networks fail in several distinguishable ways. Supports can be redundant because they share data. They can be proxy-contaminated because they measure the same target through different labels. They can be estimator-dependent because a common inference map produces agreement. They can be boundary-dependent because all sources inherit the same endpoint choice. They can be locally compatible but globally incomplete because each source removes a different subset of alternatives without closing the intersection. UEAP records these failures rather than averaging them into a single confidence label.

Theorem 10.3 (No Illicit Status-Amplification Theorem). *Many compatible supports do not form proof or final legitimacy merely by accumulation. Evidence may increase confidence inside a status class; it may promote across a status boundary only when it discharges the blocker defining the lower status. Compatible supports can therefore promote a claim only when they are target-aligned, independent in the evidence network, globally compatible, and jointly eliminate the remaining admissible alternatives or satisfy the domain’s declared evidential standard for σ .*

Proof. Compatibility alone means that evidence nodes can coexist with the claim. It does not show that they eliminate alternatives or satisfy the declared evidential standard. UEAP does not deny Bayesian updating, meta-analysis, consilience, or ordinary accumulation of independent evidence. It denies only illicit status promotion: moving from compatible or evidence-compatible status to final legitimacy when the evidence network still contains shared ancestry, unresolved alternatives, hidden common assumptions, or merely pairwise agreement. If sources share ancestry or assumptions, adding them does not increase independent constraint. If they are only pairwise compatible, a global alternative may remain. If the remaining alternatives are not eliminated or otherwise licensed by the declared status standard, the freedom exhaustion criterion fails. Therefore support can strengthen a claim only to the extent that it independently closes alternatives in the network or satisfies the declared standard for σ ; otherwise its status is evidence-compatible rather than legitimate. $\square$

Corollary 10.4 (Pairwise compatibility is not global closure). *A collection of supports can be pairwise compatible and still fail to close the claim if no global evidence network eliminates the remaining admissible alternatives.*

11 Reportability and actionability

11.1 Reportability as part of legitimacy

A claim can be fixed but not reportable in the way the author wants. A relation may be fixed without a member selection. A class may be fixed without a local coordinate. An obstruction may rule out a route without providing a positive assertion. A calibrated model may be useful but not predictive. UEAP therefore treats reportability as a final audit coordinate.

11.2 Report promotions to avoid

The most common reportability failures are status promotions. Compatible-only is reported as confirmed; calibrated-only is reported as predicted; domain-dependent is reported as universal; computation-only is reported as proven; relation-only is reported as an individual value; obstruction-only is reported as a positive construction; evidence-compatible is reported as closed. UEAP treats each as a report transformation that must be audited.

Definition 11.1 (Reportability). A claim is reportable at status s when the statement issued to users preserves target, carrier, scope, ancestry, evidence status, uncertainty, blockers, and nonclaim boundaries associated with s .

Theorem 11.2 (Report Preservation Theorem). *A fixed claim is not automatically reportable or actionable. Reporting must preserve claim status, dependencies, uncertainty, scope, and nonclaim boundaries. If the report changes any of these, the reported claim is a new claim and must be audited again.*

Proof. A report is an action on claim content. If it omits a domain boundary, upgrades calibrated-only to prediction, converts compatibility into proof, turns an obstruction into a positive assertion, or removes uncertainty needed for the status class, it changes the claim offered to the reader. That new claim has a different target, carrier, scope, or status. Therefore reportability requires preservation of the audited status. Otherwise the report is itself a laundering route. $\square$

Table 6: UEAP final status taxonomy.

Final status	Meaning
Legitimate	Target, carrier, alternatives, laundering, and reportability are closed for the requested status class σ .
Legitimate but not actionable	The claim is fixed, but no responsible operational action or report use is available.
Compatible only	The claim fits available structure, but alternatives remain.
Relation-only	A relation is fixed while individual assertions are not.
Class-level only	A class, residue, category, or quotient is fixed without selecting a member or local coordinate.
Calibrated only	The result inherits target data, proxy ancestry, or outcome-conditioned fitting.

Final status	Meaning
Domain-dependent	The claim depends on supplied boundary, scope, population, regime, or endpoint.
Evidence-compatible	Evidence agrees but does not eliminate alternatives.
Computation-only	A result was computed without claim standing or exact support.
Obstruction-only	A route or class is exactly ruled out without supporting a positive assertion.
Failed	Required standing or admissibility conditions fail.

11.3 Status precedence and blocker sets

A claim can have more than one active lower-status marker. For example, it may be both calibrated-only and domain-dependent, or relation-only and evidence-compatible. UEAP therefore does not erase coexisting blockers. The audit output is a triple

$$\text{Reg}_D(c) = (\text{Status}_D(c), \text{Blockers}_D(c), \text{NextAction}_D(c)).$$

The final reported status is the strongest status licensed by a declared precedence order, while $\text{Blockers}_D(c)$ records every active reason why stronger statuses are unavailable.

Definition 11.3 (Status precedence). A status precedence order $\preceq_\sigma$ is fixed before final assignment. Lower statuses mark stronger blockers or narrower reportability. A default UEAP order is:

$$\begin{aligned} &\text{failed} \prec \text{standing-unfixed} \prec \text{laundered/blocked} \prec \text{computation-only} \prec \\ &\text{calibrated-only} \prec \text{domain-dependent} \prec \text{evidence-compatible} \prec \text{compatible-only} \prec \\ &\text{relation/class/obstruction-only} \prec \text{legitimate-but-not-actionable} \prec \text{legitimate}. \end{aligned}$$

Domain-specific protocols may refine this order, but they must declare the refinement before status assignment.

Proposition 11.4 (Blocker preservation). *If a claim receives a higher reportable status while a lower-status blocker remains active, the report must preserve that blocker in $\text{Blockers}_D(c)$. Suppressing the blocker is report laundering.*

Proof. A blocker is an unresolved condition that prevents a stronger status from being asserted without qualification. If the blocker remains active but is omitted from the report, the report presents the claim as if the condition were discharged. That changes the report object and violates report preservation. Therefore all active blockers must remain in the registry even when a strongest reportable status is assigned. $\square$

12 Validator protocol and registry

12.1 Validator sequence

The validator is the operational product of UEAP. It is designed for use on claims from science, AI, medicine, law, engineering, policy, forecasting, finance, cosmology, and ordinary technical reasoning.

Protocol 12.1 (UEAP validator). Every audited claim is processed through the following ten questions:

1. What exactly is the claim?
2. What carrier gives the claim standing?
3. What alternatives remain admissible?
4. Which variations are skin rather than content?
5. Did the result inherit target or proxy data?
6. What scope, boundary, or operating regime is required?
7. Are evidence sources independent and globally compatible?
8. Could laundering explain the support?
9. Can the claim be reported without changing status?
10. What is the strongest defensible final status?

Step	Validator question	Typical failure status
1	What exactly is the claim?	target-unfixed
2	What carrier gives the claim standing?	carrier-unfixed
3	What alternatives remain admissible?	compatible-only
4	Which variations are skin rather than content?	meaning-unfixed
5	Did the result inherit target/proxy data?	calibrated-only
6	What scope, boundary, or operating regime is required?	domain-dependent
7	Are evidence sources independent and globally compatible?	evidence-compatible
8	Could laundering explain the support?	laundered/blocked
9	Can the claim be reported without changing status?	not reportable
10	What is the strongest defensible final status?	registry assigned

Table 7: UEAP validator sequence.

Definition 12.2 (UEAP audit basis). A UEAP audit basis is a tuple

$$\mathfrak{A} = (D, \sigma, \mathcal{C}_\sigma, \text{AdmAlt}, \sim_{\text{Skin}}, \mathcal{L}, \preceq_\sigma),$$

where D is the domain ledger, σ is the requested status class, $\mathcal{C}_\sigma$ is the carrier-standard family for that class, AdmAlt is the admissible-alternative relation, $\sim_{\text{Skin}}$ is the skin relation, $\mathcal{L}$ is the declared laundering taxonomy, and $\preceq_\sigma$ is the status-precedence order.

The certification theorem is internal to this fixed audit basis. It does not claim to exhaust every possible ordinary-language use of legitimacy; it states the possible UEAP outputs once the target domain, requested status, carrier standard, admissible alternatives, skin relation, laundering taxonomy, and precedence order have been declared.

Theorem 12.3 (UEAP Certification Theorem). *Given a fixed UEAP audit basis $\mathfrak{A}$ and a completed validator pass for claim c , the audit returns exactly one of the following:*

1. *a σ -legitimacy certificate, when target fixation, carrier adequacy for σ , alternative exhaustion modulo skin, laundering exclusion, and report preservation all hold;*

2. a lower-status report with blocker set, when the claim is fixed enough to report but one or more blockers prevent the requested status;
3. standing-unfixed, target-unfixed, carrier-unfixed, meaning-unfixed, or scope-unfixed status, when preliminary standing conditions fail;
4. audit failure due to an undeclared domain ledger, undeclared requested status, undeclared carrier-standard family, undeclared skin relation, undeclared laundering taxonomy, or undeclared status-precedence order.

No fifth validator output is available under the fixed audit basis.

Proof. The validator asks the standing questions first. If target, carrier, meaning, or scope is unfixed, the audit returns the corresponding preliminary status. If the domain ledger, requested status, carrier-standard family, skin relation, laundering taxonomy, or precedence order is undeclared, the audit cannot determine what counts as satisfying the requested status, so it returns audit failure. Assume these preliminary failures are absent. The validator then evaluates admissible alternatives, skin classification, ancestry, domain/boundary dependence, evidence-network independence, laundering routes, and report preservation. If all conditions required by σ are satisfied, the audit returns a σ -legitimacy certificate. If at least one condition required by σ fails while a weaker status remains reportable, the status-precedence order selects the strongest satisfied weaker status and the registry records all failed stronger-status conditions as blockers. These cases are exhaustive because every validator question either fixes the audit basis, fixes preliminary standing, closes a status condition, or records the condition as a blocker. Hence there is no fifth output. $\square$

12.2 Registry schema

The registry turns claim auditing into a repeatable practice. Each audited claim receives a row with fixed fields: Claim ID, claim text, target role, carrier, scope/domain, evidence/support, freedom ledger, laundering audit, ancestry graph, network audit, reportability, final status, and next action.

Theorem 12.4 (Unique Strongest Status Assignment). *For every audited claim with a completed registry row and declared precedence order $\preceq_\sigma$, UEAP assigns a unique strongest reportable status, a blocker set recording all active non-final constraints, and one next-action requirement unless the blocker set is empty.*

Proof. A completed registry row fixes the target, carrier, scope, evidence, freedom ledger, laundering audit, ancestry graph, network audit, and reportability field. The declared precedence order $\preceq_\sigma$ ranks the reportable statuses for that audit. Let S_c be the set of statuses whose conditions are satisfied by the registry row. Because the precedence order is fixed and finite on the status taxonomy used in the registry, S_c has a unique maximal element whenever it is nonempty; if it is empty, the default status is failed or standing-unfixed according to the first failed coordinate. The blocker set is the set of unsatisfied conditions required for every stronger status above that maximal element. The next action is the minimal declared operation that would discharge at least one load-bearing blocker: target fixation, carrier replacement, alternative exhaustion, ancestry separation, boundary declaration, evidence-network repair, laundering discharge, or report narrowing. Therefore the completed row determines one strongest reportable status and preserves all active blockers. $\square$

12.3 Registry discipline

The registry is not merely archival. It prevents the same claim from being silently reclassified across contexts. The Claim ID fixes the audit object. The target and carrier fields prevent standing drift. The freedom ledger records what remains open. The laundering audit records invalid promotion routes. The reportability field records the exact wording or status class that may be used downstream. The next-action field prevents a failed or partial audit from becoming a dead end: it states what would be needed to strengthen or close the claim.

13 Relation to existing practices and neighboring literatures

13.1 Construct and measurement validity

Construct-validity and measurement-validity work ask whether an instrument or proxy actually measures the intended target [1]. Scientific measurement, theory comparison, and model assessment also rely on explicit uncertainty, robustness, and validation practices [3, 4]. UEAP generalizes that question by separating the target role, carrier role, ancestry graph, and report object. It does not replace measurement validity; it records which status remains after measurement, proxy, and reportability blockers are preserved. A construct-validity failure becomes one normal form among a broader set of claim-standing failures. UEAP’s added contribution is blocker-preserving status assignment: a measurement may be compatible, calibrated, domain-dependent, or legitimate for a restricted target without being promoted to a stronger claim.

13.2 Causal inference and proxy leakage

Causal inference distinguishes association, identification, intervention, and counterfactual support [2]. UEAP does not replace those tools. It treats them as carrier standards for causal-status claims. Its contribution is to expose proxy laundering and target ancestry even outside explicitly causal models, including benchmarks, surrogate endpoints, model calibration, and forensic reconstruction. A variable can support prediction, calibration, explanation, or causal inference only at the status class whose carrier conditions it satisfies.

13.3 Model validation, uncertainty quantification, and robustness

Verification, validation, uncertainty quantification, and sensitivity analysis examine how models behave under numerical, physical, and assumption perturbations [4, 3]. UEAP turns these practices into status coordinates: robustness belongs to the domain ledger, evidence network, freedom ledger, or report object, depending on what it affects. In clinical and biomedical settings, analogous distinctions arise in prediction-model reporting and surrogate-endpoint assessment [7, 8]. Fit, calibration, uncertainty, and domain dependence are not collapsed into one confidence score; they are assigned distinct blockers and reportable statuses.

13.4 Formal verification and proof checking

Formal verification and proof checking can be standing-bearing carriers when their target, input ledger, inference rules, and report class are fixed. This is compatible with ordinary verification practice: UEAP

asks what claim a verified artifact is allowed to carry, not whether verification is valuable. UEAP’s distinction is between trace conformity and claim standing. A proof assistant result, solver output, or derivation trace may certify a syntactic or formal status, but it supports a stronger mathematical or applied claim only when the carrier is adequate for that requested status class. This is why computation alone is not legitimacy, while computation within an adequate carrier can be legitimacy-relevant.

13.5 Evidence law and forensic admissibility

Evidence law and forensic reconstruction already distinguish relevance, admissibility, weight, chain of custody, expert qualification, and permissible use; Daubert-style reliability tests and forensic-science reform reports are examples of domain standards that would enter UEAP through the carrier and domain ledgers [9, 10]. UEAP abstracts this into the language of target, carrier, evidence network, laundering, and reportability. It does not replace legal doctrine. It supplies a general status logic for preventing compatibility, authority, or narrative coherence from being laundered into causation or proof beyond the carrier’s standing.

13.6 AI and machine-learning benchmark governance

AI and machine-learning evaluation expose the same structure in a modern form: benchmark leakage, proxy targets, train/test contamination, metric gaming, benchmark-specific overfitting, model-card ambiguity, and capability over-reporting [5, 6]. UEAP treats these not as miscellaneous warnings but as ledgered blockers: ancestry failure, proxy laundering, metric/baseline laundering, evidence-network failure, domain dependence, and report promotion.

13.7 What UEAP adds

The neighboring literatures are mature and domain-specific, including proper scoring and forecast-evaluation work in forecasting [12] and systems-engineering safety and verification practice [11]. UEAP’s contribution is not to replace them with new vocabulary. It gives a cross-domain status grammar: the audit records the target, carrier, freedoms, ancestry, domain, evidence network, laundering routes, report object, blocker set, and next action in one registry row. This permits heterogeneous claims to be compared without pretending that they share the same evidence carrier.

14 Pilot audits: demonstration status

The pilots below are not offered as independent empirical validation of UEAP and are not used as evidence that the protocol is correct. They are worked classification examples showing how the registry handles different claim forms whose detailed source analyses are supplied elsewhere. The source-specific mathematical or empirical claims remain conditional on the cited source analyses. Their function here is operational: each pilot exhibits target fixation, carrier standing, active freedoms, laundering risks, final status, blockers, and reportable form in a different domain.

15 Pilot Audit I: canonical worked audit – Black–Scholes valuation

15.1 Claim under audit

The first pilot claim, based on the Black–Scholes rigidity manuscript and the classical Black–Scholes–Merton setting [15, 18, 19], is:

Within the fixed one-factor representative-invariant local valuation interface, the Black–Scholes operator is the unique nondegenerate local valuation operator.

This is not an empirical claim that option markets are lognormal. It is a claim about claim identity, representative invariance, and same-scope valuation.

15.2 UEAP mapping

UEAP object	Black–Scholes audit instance
Target	Representative-independent value of a terminal payoff claim.
Carrier	Payoff quotient $D_\Phi/\sim$, one-factor local state (S, t) , self-financing redescription relation, terminal trace, and fixed numeraire.
Freedom ledger	Payoff identity, hidden selector, local state, financing relation, terminal trace, local continuation law, scale/time symmetry, numeraire.
Skin	Payoff-preserving lawful coordinate, hedge-accounting, or self-financing redescription that does not affect value.
Laundrying risk	Treating stochastic representation, calibration surface, martingale-measure selector, or implied-volatility fit as claim identity.
Reportability	Structural operator-rigidity claim within the fixed interface; richer empirical models are controlled scope changes.

Table 8: UEAP mapping for the Black–Scholes pilot.

15.3 Detailed registry analysis

The Black–Scholes pilot is the cleanest example because the target is not an empirical market fact but an operator-classification claim. The target role is the representative-independent local value of a terminal payoff quotient. The carrier consists of the payoff quotient, the self-financing redescription relation, the one-dimensional local state, deterministic pointwise valuation, terminal trace, scale covariance, time homogeneity, and fixed numeraire.

The freedom ledger is explicit. A change in quotient identity makes the valued object different. A hidden measure selector makes the value depend on a chosen probability law rather than claim identity. A local-volatility surface introduces additional state or coefficient data. Transaction costs and funding frictions change the self-financing carrier. Jumps and memory kernels change the continuation law. Time-dependent volatility changes time homogeneity. Stochastic rates change the numeraire/state structure. None of these moves is illegitimate; each changes the domain of the valuation claim.

The laundrying routes are also explicit. Expectation agreement can mask representative dependence; calibration can select a surface after seeing prices; implied volatility can be mistaken for a physical volatility claim; a stochastic representation can be mistaken for the operator itself. UEAP’s report

object prevents these promotions by stating the result as fixed-interface rigidity rather than empirical market ontology.

15.4 Registry-style audit table

Registry field	Black–Scholes audit entry
Literal claim	Black–Scholes is the unique nondegenerate local valuation operator in the fixed representative-invariant one-factor interface.
Requested status	Structural legitimacy within a declared valuation interface.
Target	Claim-value operator, not empirical asset dynamics.
Carrier	Payoff quotient, self-financing redescription, local state, terminal trace, scale/time symmetry, fixed numeraire.
Active freedoms	Hidden selector, state enlargement, financing change, terminal/boundary addition, continuation-law change, numeraire change.
Discharged freedoms	Pure payoff-preserving redescription is skin; all non-Black–Scholes rivals are located in declared departure loci.
Laundering routes excluded	Expectation masking, implied-volatility ontology, calibration-surface selection, martingale-measure selector.
Remaining boundary	Fixed one-factor local frictionless interface only.
Final status	Legitimate within declared fixed interface.
Reportable formulation	Operator-rigidity theorem, not empirical lognormal-market claim.

Table 9: Registry-style UEAP audit for the Black–Scholes pilot.

15.5 Audit result

The valuation paper first fixes claim identity by a quotient of descriptions under payoff-preserving, self-financing, selector-free redescription. A valuation is representative-invariant exactly when it factors through that quotient. Hidden data such as calibration surfaces, probability-law selections in incomplete markets, or representative-selection conventions are excluded unless declared as part of a richer valuation problem.

The same paper then classifies departures from the fixed interface. A non-Black–Scholes rival must change the payoff identity quotient, use a hidden selector, enlarge the local state, change self-financing or payoff-linearity, add terminal or boundary data, replace the local continuation law, or break scale covariance, time homogeneity, or the fixed numeraire. Each route is bookkeeping, a declared scope change, or a representative-invariance failure.

Audit 15.1 (Black–Scholes UEAP status). The claim receives the status *legitimate within declared fixed interface*. Its reportable form is: within the payoff-linear, deterministic, one-factor, local, continuous hedge-neutral, self-financing, scale-covariant, time-homogeneous valuation class with fixed risk-free numeraire and terminal trace, the Black–Scholes operator is the unique nondegenerate representative-invariant local operator. The result is not reportable as an empirical assertion that markets satisfy constant-volatility lognormal dynamics.

Attempted support or variation	UEAP classification	Reportable consequence
Implied volatility surface	Domain/state enrichment or calibration object	Useful empirical representation, not same-scope counterexample.
Alternative martingale measure in incomplete market	Hidden selector unless declared as market data	Compatible pricing family, not quotient-fixed value.
Stochastic volatility or jumps	State or continuation-law change	Richer model, not same fixed interface.
Transaction costs or funding frictions	Self-financing/linearity change	Different valuation problem.
Pure payoff-preserving redescription	Skin	No change in value.

Table 10: Selected UEAP classifications in the Black–Scholes pilot.

16 Pilot Audit II: formal extension audit – self-adjoint extensions

16.1 Claim under audit

The second pilot claim, based on the self-adjoint extension manuscript and standard operator theory [16, 20, 21, 22], is:

A particular self-adjoint extension is canonically selected by the original symmetric operator base.

This is a formal pilot for carrier standing and auxiliary-data exposure. It demonstrates that extension is not automatically same-base legitimacy.

16.2 UEAP mapping

UEAP object	Self-adjoint extension audit instance
Target	Selection of a self-adjoint realization for a densely defined symmetric operator.
Carrier	Original symmetric operator base, deficiency spaces, extension parameter, boundary data, or form-order endpoint.
Freedom ledger	Carrier change, old graph alteration, boundary condition, deficiency-space unitary, phase choice, normalization, variational criterion, endpoint selection.
Skin	Unitary redescription preserving the fixed operator base and extension data.
Laundering risk	Treating a selected extension as internal when the selector is boundary, phase, endpoint, or physical/evaluative data not fixed by the original base.
Reportability	Essential self-adjointness is conservative; Friedrichs/Krein selection is endpoint-criterion dependent; arbitrary extension is externally selected or framework-changing.

Table 11: UEAP mapping for the self-adjoint extension pilot.

16.3 Detailed registry analysis

The self-adjoint extension pilot separates three statuses that are often compressed. First, if the operator is essentially self-adjoint, the closure is fixed by the original graph and Hilbert-space structure; the extension behaves as a conservative limiting case. Second, if multiple extensions exist and no endpoint or boundary criterion is declared, the carrier is incomplete for unique selection. Third, if a closed-form endpoint criterion is explicitly part of the carrier, the Friedrichs or Krein selection can be legitimate under that declared criterion.

The freedom ledger identifies what can move: the operator graph, domain, deficiency-space unitary, scalar phase, boundary condition, form order, comparison criterion, or physical/evaluative selection principle. A claim that says “the operator selects the extension” while using one of those data is laundered unless the datum is already standing-bearing in the base.

This pilot shows that a mathematically canonical construction may still be domain-dependent. Canonical relative to an enriched carrier is not identical to internal relative to the old base. UEAP preserves that distinction without denying the legitimacy of the enriched construction.

16.4 Registry-style audit table

Registry field	Self-adjoint extension audit entry
Literal claim	The original symmetric operator base canonically selects a particular self-adjoint extension.
Requested status	Unique internal selection of a self-adjoint realization.
Target	Self-adjoint extension selection.
Carrier	Original operator graph, Hilbert space, deficiency spaces, admissibility policy, endpoint/boundary data if declared.
Active freedoms	Deficiency-space unitary, scalar phase, boundary condition, endpoint criterion, physical/evaluative selector.
Discharged freedoms	Essential self-adjointness discharges extension choice; Friedrichs/Krein discharge by explicit endpoint criterion.
Laundering routes excluded	Treating auxiliary boundary or phase data as if generated by the original base.
Remaining boundary	Nonzero-deficiency case requires declared selector or changed operator framework.
Final status	Carrier-unfixed or failed for blanket claim; legitimate only under essential self-adjointness or declared endpoint criterion.
Reportable formulation	Extension selection is conservative, endpoint-conditioned, externally selected, or framework-changing.

Table 12: Registry-style UEAP audit for the self-adjoint extension pilot.

16.5 Audit result

The operator-theoretic source gives a fixed-base classification. An extension or readback procedure is conservative on old-base-relevant content, depends on auxiliary data not fixed by the base, or changes operator graph, carrier, language, or admissibility policy. In the self-adjoint case, von Neumann deficiency theory parameterizes extensions by unitaries $U : N_+ \rightarrow N_-$ when deficiency indices agree. Redescription-invariant candidates must be unitary intertwiners between the deficiency representations.

If no intertwiner exists, no invariant internal selection exists. If one exists, its scalar phase orbit gives distinct self-adjoint extensions, so invariance alone still does not select a unique nontrivial realization.

The positive endpoint case is different. For a lower semibounded symmetric operator, the Friedrichs extension is the unique upper endpoint in the closed quadratic-form order. In the nonnegative case, the Krein–von Neumann extension is the corresponding lower endpoint. These are not hidden selections; they are legitimate only when the endpoint criterion is part of the base.

Audit 16.1 (Self-adjoint extension UEAP status). The blanket claim that the original symmetric operator base canonically selects a unique self-adjoint extension is *failed* or *carrier-unfixed* in the nonzero-deficiency case. Essential self-adjointness receives a conservative legitimate status. Friedrichs selection receives legitimate status under a declared form-dominance endpoint criterion. Krein–von Neumann selection receives legitimate status under the dual endpoint criterion in the nonnegative case. Arbitrary extension selection is domain-dependent or externally selected unless its boundary, phase, normalization, comparison, variational, or physical/evaluative datum is explicitly included in the carrier.

17 Pilot Audit III: reportability audit – cosmological transport closure

17.1 Claim under audit

The third pilot claim, based on the cosmological transport-closure manuscript and its public-data stack context [17, 23, 24, 25, 26], is:

The SH0ES-scale Hubble value is realized as a same-parameter transport of the early-universe anchor under the implemented transport-closure constraints.

This pilot is empirical and numerical. Its UEAP role is not to resolve cosmology, but to show reportability discipline under a constrained inference architecture.

17.2 UEAP mapping

UEAP object	Cosmological transport-closure audit instance
Target	Same-parameter early-to-late transport of H_0 .
Carrier	Early anchor, transport map, quotient observables, continuation structure, boundary/calibration ledger, estimator declaration, hidden-remainder test.
Freedom ledger	Anchor split, tensor omission, quotient mismatch, continuation patch, boundary repair, estimator selector, hidden remainder.
Skin	Numerically different presentation that preserves all seven same-domain type coordinates.
Laundering risk	Treating calibration shift, fitted transport correction, hidden parameter, or post-hoc patch as a same-domain resolution.
Reportability	Conditional non-resolution within implemented basis and likelihood configuration; no rejection of SH0ES, no global model-selection claim, no exclusion of all new physics.

Table 13: UEAP mapping for the cosmological transport pilot.

17.3 Detailed registry analysis

The cosmological pilot is primarily a reportability audit. The cited transport-closure manuscript deliberately avoids stronger claims that the retained computation does not support. It does not claim a full click/clippy Planck production result. It does not claim a Bayes-factor or AIC/BIC comparison against an independently sampled unrestricted Λ CDM baseline. It does not reject the SH0ES data. It does not exclude new physics outside the tested basis. Its report object is narrower: under a declared transport-closure penalty and low-dimensional correction basis, the posterior remains near the Planck/BAO regime and does not activate a SH0ES-scale same-parameter transport correction.

In UEAP terms, this is a successful non-promotion. The computation is not laundered into a universal cosmological conclusion. The boundary ledger remains visible: native Planck validation configuration, DESI DR1 BAO, Pantheon+/SH0ES treatment, penalty weight, transport basis, retained chain ensemble, and convergence diagnostics. The final status is conditional non-resolution, not final disproof or model selection.

17.4 Registry-style audit table

Registry field	Cosmological transport audit entry
Literal claim	SH0ES-scale H_0 is realized as same-parameter transport under the implemented constraints.
Requested status	Same-domain numerical transport realization.
Target	Early-to-late parameter identity for H_0 .
Carrier	Anchor, transport map, observable quotient, continuation, boundary ledger, estimator declaration, hidden-remainder diagnostic.
Active freedoms	Anchor split, tensor omission, quotient mismatch, continuation patch, boundary repair, estimator selector, hidden remainder.
Discharged freedoms	Declared transport basis and violation functional test the constrained sector; retained chains support posterior-location claim only.
Laundering routes excluded	Calibration shift as resolution, hidden fitted correction, post-hoc transport patch, over-reporting as SH0ES rejection.
Remaining boundary	Native Planck validation configuration, retained transport basis, penalty range, no unrestricted matched baseline model-selection claim.
Final status	Conditional non-resolution under implemented basis.
Reportable formulation	SH0ES-scale same-parameter shift is not realized in the retained constrained posterior; no SH0ES rejection and no global new-physics exclusion.

Table 14: Registry-style UEAP audit for the cosmological transport pilot.

17.5 Numerical audit summary

The transport-closure manuscript defines a numerical cosmological inference tuple and seven same-domain type coordinates: anchor A , tensor witness W , quotient-observable class Q , continuation C , boundary-record ledger B , estimator E , and hidden remainder M . The AASC violation functional is the weighted sum of residuals in these coordinates. The admissible numerical solution space is the zero set of the residual map.

The retained five-chain constrained posterior, using the native Planck validation likelihood configuration

with DESI DR1 BAO and Pantheon+/SH0ES information, reports

$$H_0^{\text{AASC}} = 69.208 \pm 0.352 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Delta T(0) = 0.0267 \pm 0.0588 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The reported $\Delta T(0)$ is statistically consistent with zero in the tested basis. The distance from the constrained posterior mean to the SH0ES central value is approximately 3.49σ under the comparison stated in the cited transport-closure manuscript.

The graphical diagnostics in the cited transport-closure manuscript are reportability-relevant. The constrained H_0^{AASC} posterior is centered near 69.2 while SH0ES is marked near 73; the present-epoch transport correction is concentrated near zero; and the penalty-weight sensitivity checks over the retained $10^5, 10^6, 10^7$ range do not migrate the constrained posterior toward the SH0ES regime.

Audit 17.1 (Cosmology UEAP status). The claim that the SH0ES-scale value is realized as a same-parameter transport under the implemented constraints receives the status *conditional non-resolution*. The reportable claim is: within the implemented transport basis, violation functional, native Planck validation configuration, and retained posterior ensemble, the SH0ES-scale shift is not realized as an admissible same-scope transport correction. This is not reportable as a rejection of SH0ES, a model-selection result against unrestricted Λ CDM, or an exclusion of all possible new-physics extensions.

18 Discussion

18.1 Why UEAP is not merely a checklist

UEAP has checklist outputs, but its logic is not checklist-like. The validator follows a dependency order forced by the roles of claim standing. Target and carrier come before alternatives. Alternatives come before laundering. Laundering comes before reportability. Reportability comes before actionability. A domain expert may supply the content of the carrier, but the status logic is stable across domains.

The nontrivial content is sixfold. First, UEAP separates standing from truth: a true-looking output can remain standing-unfixed. Second, it separates carrier adequacy from evidence quantity: more evidence cannot repair the wrong carrier. Third, it treats laundering as a formal status-preserving failure rather than as rhetoric. Fourth, it preserves weaker statuses rather than forcing binary acceptance or rejection. Fifth, it uses one audit grammar across formal, empirical, computational, and institutional claims. Sixth, it produces registry rows that preserve blockers and next actions, preventing later status inflation.

18.2 Synthesis with existing practices

The related literatures supply domain-specific carrier standards. UEAP supplies the cross-domain status logic that prevents those standards from being silently exceeded. Its role is therefore integrative: it does not replace construct validity, causal identification, verification, evidence law, or benchmark governance; it records when those practices support the requested status, when they support only a weaker status, and which blockers remain.

18.3 Why the pilots are deliberately heterogeneous

The three pilots show different UEAP strengths. Black–Scholes shows quotient identity and representative invariance. Self-adjoint extension shows auxiliary-data and endpoint selection discipline.

Cosmological transport closure shows constrained numerical reportability and non-overclaiming. Their shared structure is target, carrier, freedom, laundering, and reportability.

18.4 Scope of the present contribution

The present contribution is a protocol paper. It does not require the full AASC corpus as a prerequisite, but it makes the kernel visible enough to explain why the protocol is not arbitrary. The modules developed here—laundering taxonomy, ancestry graphs, domain closure, evidence networks, validator registry, and domain instantiation guides—are presented as operational audit components rather than as substitutes for domain-specific standards.

19 Conclusion

UEAP generalizes exactness into a domain-general protocol for claim-standing and reportability assessment. A claim becomes legitimate at a requested status class only when its target and carrier are fixed for that class, admissible alternatives are exhausted or reduced to skin, laundering routes are blocked, and the final report preserves the audited status. Exact numerical prediction is one high-strength output, not the master category.

The protocol assigns the strongest defensible status rather than forcing premature closure. It can report legitimate, legitimate-but-not-actionable, compatible-only, relation-only, class-level-only, calibrated-only, domain-dependent, evidence-compatible, computation-only, obstruction-only, or failed. The validator and registry turn this into a repeatable practice. The worked audits demonstrate that the same audit roles recur in valuation, operator extension, and numerical cosmological inference.

The final practical rule is simple: no claim should be reported at a strength greater than its target fixation, carrier standing, freedom exhaustion, laundering audit, and reportability permit.

A Formal dependency map

Table 15: UEAP dependency map.

UEAP result	Depends on	Main blocker closed
Pre-Audit Coordinate Exhaustion	Pre-audit observation frame	missing coordinate objection
Claim Standing Necessity	Kernel bridge: reference and standing	target/carrier drift
Freedom Exhaustion Criterion	Exactness arc generalization and skin quotient	unresolved alternatives
Laundering Exclusion	Irreversibility/no-repair discipline	status promotion by transformation
Calibration-Only Theorem	Ancestry graph and target/proxy deletion test	target leakage
Boundary Declaration Principle	Domain ledger	endpoint and scope laundering

UEAP result	Depends on	Main blocker closed
No Illicit Status-Amplification Theorem	Evidence network, declared evidential standard, and alternative exhaustion	illegitimate support aggregation
Report Preservation Theorem	Report object and status taxonomy	over-reporting
UEAP Certification Theorem	Fixed audit basis and validator sequence	uncontrolled validator outcomes
Unique Strongest Status Assignment	Validator sequence and registry fields	ambiguous final status

B Registry template

Table 16: UEAP registry schema.

Field	Purpose
Claim ID	Stable identifier for the audited claim.
Claim text	Literal claim under audit.
Target role	Object, relation, event, status, class, or value asserted.
Carrier	Structure that supports standing.
Scope/domain	Domain, boundary, population, regime, metric, and operating envelope.
Evidence/support	Sources, arguments, models, data, proofs, or computations used.
Freedom ledger	Active and closed alternatives.
Laundering audit	Detected or excluded laundering routes.
Ancestry graph	Target/proxy/fitting dependencies.
Network audit	Evidence independence, overlap, and global compatibility.
Reportability	How the claim may be responsibly stated.
Final status	Strongest defensible status.
Next action	Certificate, input, closure, or correction required to strengthen the claim.

C Completed pilot registry rows

Table 17: Completed pilot registry rows, compressed.

Claim ID	Claim text	Final status	Next action/report boundary
UEAP-BS-01	Black–Scholes is the unique nondegenerate local valuation operator in the fixed representative-invariant one-factor interface.	Legitimate within declared fixed interface.	Do not report as empirical lognormal-market claim; richer models are scope changes.
UEAP-SA-01	A particular self-adjoint extension is canonically selected by the original symmetric operator base.	Failed or carrier-unfixed in nonzero-deficiency case; legitimate only under essential self-adjointness or declared endpoint criterion.	Declare boundary, phase, endpoint, variational, or physical/evaluative selector.
UEAP-COS-01	SH0ES-scale H_0 is realized as same-parameter transport under the implemented transport-closure constraints.	Conditional non-resolution.	Report only relative to implemented basis, likelihood configuration, violation functional, and retained posterior; no SH0ES rejection or global model-selection claim.

D Expanded laundering table

Table 18: Expanded laundering audit table.

Route	Typical signal	Why it blocks legitimacy	Possible status
Semantic	Same words, changed meaning	Target is not fixed	meaning-unfixed
Proxy	Support variable reconstructs target	Independence is illusory	calibrated-only
Calibration	Fit reported as prediction	Target ancestry remains	calibrated-only
Domain	Scope changes after support known	Domain ledger is hidden	domain-dependent
Endpoint	Boundary selected for desired result	Endpoint freedom remains	domain-dependent
Evidence network	Many agreeing sources share ancestry	No independent elimination	evidence-compatible

Route	Typical signal	Why it blocks legitimacy	Possible status
Computation	Solver output substitutes for carrier	Computation alone lacks target/carrier audit	computation-only
Authority	Consensus or institutional status replaces support	Authority alone is not the required carrier unless authority is the audited target	carrier-unfixed
Post-hoc explanation	Explanation created after target observation	Predictive ancestry is missing	calibrated-only
Metric/baseline	Score or unit choice manufactures result	Baseline freedom remains	domain-dependent
Target drift	Target moves after evidence is known	Reference fails	target-unfixed

E Coordinate-completeness table

Table 19: Why the UEAP coordinate list is complete for status-relevant variation.

Coordinate	What it captures	Typical failure
Target T	What is asserted: object, relation, value, event, class, or status.	target drift
Carrier C	What bears support for the requested status.	carrier-unfixed
Semantic content S	Meaning under wording, interpretation, and redescription.	meaning-unfixed
Representation R	Which presentations count as the same claim.	representative dependence
Ancestry A	Calibration, proxy, fitting, tuning, and target-derived support paths.	calibrated-only
Domain ledger D	Scope, boundary, endpoint, regime, metric, baseline, and population.	domain-dependent
Evidence network E	Support nodes, independence, common ancestry, and global compatibility.	evidence-compatible
Report/use profile P	Wording, actionability, visualization, decision use, downstream uptake.	not reportable
Requested status U	The strength class being asserted.	status-class mismatch

Status of the domain instantiation guides. The domain instantiation guides that follow are not comprehensive literature reviews and not substitutes for field-specific standards. They do not specify domain standards; they specify how externally supplied domain standards enter the UEAP registry. Each guide identifies how the protocol’s audit coordinates, blocker classes, laundering routes, and reportability rules apply in a domain. Domain-specific evidential, legal, regulatory, clinical, engineering,

financial, or institutional standards must be supplied through the carrier, domain ledger, and requested status class of the relevant audit. The guides therefore specify how UEAP classifies claim standing and reportability once those standards are declared; they do not provide medical, legal, engineering, financial, regulatory, or clinical advice.

F Science and theory-testing domain instantiation guide

F.1 Domain scope

This domain instantiation guide applies UEAP to scientific claims about empirical phenomena, mechanisms, parameters, model comparison, numerical simulation, anomaly resolution, and theory confirmation. It covers predictions, fitted models, parameter estimates, explanatory mechanisms, empirical confirmations, computational results, and theory-comparison statements. It does not adjudicate sociological priority, journal prestige, institutional consensus, or metaphysical speculation unless those are themselves the declared targets of the claim.

A scientific claim is UEAP-auditable only when the target phenomenon, the support carrier, the domain of application, and the evidential role are fixed. A mathematical derivation, experimental protocol, simulation, fitted model, or data set can all be carriers, but each carries only the status for which it is adequate. A fitted curve may support compatibility; it does not by itself support prediction. A simulation may support computation-only until its input ledger, numerical method, boundary conditions, validation data, and uncertainty discipline are fixed. A parameter estimate may support a domain-dependent inference while failing to support a universal theory claim.

F.2 Typical scientific claim forms

Table 20: Scientific claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Prediction	Model M predicts value x before observation.	Calibration laundering; target ancestry.
Explanation	Mechanism A explains phenomenon B .	Post-hoc explanation laundering.
Confirmation	Evidence E confirms theory T .	Evidence-network laundering.
Parameter claim	Parameter θ has value $v \pm e$.	Estimator, prior, unit, and baseline freedom.
Theory comparison	Theory T_1 outperforms T_2 .	Metric/baseline laundering.
Anomaly resolution	Model M resolves anomaly A .	Domain laundering; endpoint repair; hidden remainder.
Mechanism claim	Variable X drives phenomenon Y .	Proxy laundering; causal-carrier inadequacy.
Simulation claim	Solver output demonstrates property P .	Computation laundering; boundary freedom.

F.3 Standing-bearing carriers in science

Scientific carriers include deductive derivations, experimental protocols, validated measurement instruments, statistical models with declared estimators, simulations with validated inputs and boundary conditions, causal or mechanistic models, constraint-exhaustion proofs, and reproducible data pipelines. The carrier must be adequate for the requested status class. For example:

- A controlled experiment may support a causal claim only under the declared intervention, measurement, and confounding discipline.
- A fitted model may support interpolation or compatibility, but not target-independent prediction unless calibration ancestry is separated.
- A theory comparison requires a declared comparison class, metric, baseline, and admissible alternative space.
- A numerical simulation requires a solver ledger, discretization or approximation ledger, validation evidence, and boundary-condition record before it can support more than computation-only status.
- A parameter claim requires estimator, likelihood or error model, prior or regularization discipline where relevant, units, calibration, and uncertainty report.

F.4 Required scientific domain ledger

Table 21: Scientific domain ledger.

Ledger field	Required audit detail
Target phenomenon	What value, relation, mechanism, prediction, or anomaly is asserted?
Model class	Which hypotheses, equations, parameters, and alternatives are allowed?
Measurement regime	Which instruments, protocols, calibrations, and uncertainty models generated the evidence?
Estimator	Which inference map produced the reported result?
Boundary conditions	Which initial, endpoint, or domain assumptions are supplied?
Calibration ancestry	Was the result fitted, trained, tuned, or selected using target data?
Error model	What uncertainty discipline applies, and what does the reported interval mean?
Comparison class	Against which alternatives is the claim being made?
Report object	Is the output a prediction, explanation, compatibility result, obstruction, or domain-dependent statement?

F.5 Common scientific freedoms

The common active freedoms are target freedom, parameterization freedom, model-class freedom, estimator freedom, measurement-calibration freedom, endpoint/boundary freedom, scale/unit freedom, anomaly-selection freedom, comparison-metric freedom, and reportability freedom. Each freedom is

harmless only if it is skin, declared input, inadmissible, obstruction-only, or uniquely fixed. Otherwise it blocks final scientific legitimacy at the requested status.

F.6 Scientific laundering routes

Table 22: Scientific laundering routes.

Route	Scientific manifestation	UEAP blocker
Calibration laundering	Fitted output reported as prediction.	calibrated-only
Proxy laundering	Surrogate or proxy variable treated as the target.	target/proxy ancestry blocker
Endpoint laundering	Boundary or initial condition selected to recover result.	domain-dependent
Computation laundering	Simulation output treated as legitimacy without validation.	computation-only
Metric laundering	Scoring rule or baseline selected after outcome.	metric/baseline blocker
Authority laundering	Consensus, prestige, or citation count substitutes for carrier.	authority-only/carrier-unfixed
Post-hoc explanation	Explanation after observation reported as prediction.	calibrated-only/post-hoc
Theory-choice laundering	Alternative class is narrowed after the result is known.	comparison-class blocker

F.7 Scientific status assignment rules

A scientific claim receives legitimate status only if target, carrier, alternatives, ancestry, domain, evidence network, and report object are closed for the requested status. It is compatible-only if it fits the available structure while alternatives remain. It is calibrated-only if fitted to the target or proxy. It is domain-dependent if it depends on supplied boundary, scope, measurement, or operating regime. It is computation-only if generated by simulation without standing-bearing validation. It is obstruction-only if a model class is ruled out without positive replacement. It is evidence-compatible if evidence agrees but does not eliminate alternatives.

F.8 Miniature worked audit

Claim: “A fitted cosmological extension resolves the Hubble tension.” The target is same-parameter early-to-late resolution. The carrier is a model, likelihood stack, transport map, and boundary ledger. Active freedoms include calibration ancestry, estimator choice, boundary record, transport basis, and hidden tensor witness. A fit to the late value without transport closure is calibration or domain repair, not same-domain resolution. The strongest status without transport residual closure is domain-dependent or calibrated-only non-resolution. Final status: conditional non-resolution / domain-dependent constrained result. Next action: supply a transport certificate, hidden-remainder test, and matched baseline comparison.

F.9 Science guide nonclaims

UEAP does not replace experimentation, statistics, peer review, or scientific expertise. It classifies the strongest status a scientific claim can responsibly bear and records the blockers that prevent stronger report.

G AI/ML benchmark and capability domain instantiation guide

G.1 Domain scope

This domain instantiation guide applies UEAP to machine-learning and AI claims about benchmark performance, model capability, alignment or safety evaluation, interpretability, robustness, generalization, training-data leakage, tool use, and deployment readiness. It excludes product marketing, aesthetic preference, anthropomorphic speculation, and broad claims about intelligence unless the target, carrier, and evaluation standard are declared.

The core AI/ML distinction is between a benchmark result and the capability reported from it. A model score can have standing as a benchmark score while lacking standing as a general capability claim. A demo can illustrate behavior while failing to support robustness. A red-team result can support a threat-model-relative safety status while failing to support deployment safety in general.

G.2 Typical AI/ML claim forms

Table 23: AI/ML claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Benchmark	Model M scores S on benchmark B .	contamination; metric freedom
Capability	Model M can reason mathematically.	target drift; benchmark proxy
Generalization	Performance transfers to real tasks.	domain laundering
Safety	Model M is safe or aligned.	reportability overreach
Interpretability	Feature F explains behavior.	proxy explanation
Robustness	Model resists adversarial prompts.	threat-model scope drift
Deployment readiness	System is ready for domain D .	operating-regime and evidence-network gaps

G.3 Standing-bearing carriers in AI/ML

AI/ML carriers include held-out benchmarks with provenance, contamination audits, pre-registered evaluation protocols, adversarial test suites, ablation studies, causal interventions on model behavior, deployment-specific risk evaluations, reproducible evaluation harnesses, model/data documentation, and monitoring evidence. The carrier must match the requested status. A benchmark score supports benchmark performance, not necessarily broad capability. A model card supports disclosure, not safety.

A red-team result supports local robustness only under the declared threat model. A curated demo supports example performance, not general competence.

G.4 Required AI/ML domain ledger

Table 24: AI/ML domain ledger.

Ledger field	Required audit detail
Model identity	Version, checkpoint, training regime, tool access, inference settings.
Data provenance	Train/test separation, contamination audit, benchmark exposure, synthetic data use.
Benchmark target	What the benchmark measures and what it does not measure.
Evaluation protocol	Prompting, sampling, temperature, tools, scoring, retries, human grading.
Comparator	Baselines, selection criteria, and whether comparison was pre-specified.
Deployment scope	Intended task environment, users, constraints, and failure costs.
Threat model	Adversary class, attack surface, allowed tools, and robustness boundary.
Report object	Score, capability, reliability, safety, interpretability, or deployment readiness.

G.5 Common AI/ML freedoms

Common freedoms include benchmark target freedom, contamination freedom, prompt/protocol freedom, metric freedom, sampling freedom, tool-use freedom, deployment-domain freedom, capability-label freedom, evaluator freedom, and reportability freedom. A score is not a capability unless these freedoms are fixed or classified. A benchmark can be legitimate while the capability report remains target-unfixed.

G.6 AI/ML laundering routes

Table 25: AI/ML laundering routes.

Route	AI/ML manifestation	Blocker
Benchmark laundering	Benchmark score treated as broad intelligence or reliability.	target drift
Contamination laundering	Leaked or trained-on data treated as generalization.	calibrated-only
Prompt laundering	Tuned prompt or scaffold treated as base-model capability.	carrier drift
Metric laundering	Favorable metric selected after outcome.	metric blocker

Route	AI/ML manifestation	Blocker
Demo laundering	Curated example treated as robust ability.	evidence-compatible only
Authority laundering	Lab status or leaderboard rank substitutes for audit.	authority-only
Safety laundering	Narrow eval reported as broad safety.	reportability failure
Interpretability laundering	Proxy explanation treated as mechanism.	proxy blocker
Threat-model laundering	Unstated adversary boundary supports robustness claim.	domain-dependent

G.7 AI/ML status assignment rules

A legitimate benchmark claim requires fixed benchmark target, contamination exclusion or declaration, fixed protocol, reproducible score, and report-preserving wording. A capability claim requires a carrier adequate for the claimed capability rather than merely a benchmark proxy. A safety claim requires a declared threat model, test coverage, residual-risk report, and deployment scope. A model with examples but no target-aligned validation receives evidence-compatible or compatible-only status. A score produced under hidden contamination receives calibrated-only or failed status.

G.8 Miniature worked audit

Claim: “Model M is reliable at legal reasoning because it scores highly on benchmark B .” The target is deployment-level legal reasoning reliability. The carrier is benchmark B . Active freedoms include benchmark target, contamination, task distribution, legal-domain coverage, prompt protocol, tool use, and deployment context. If B tests answer recall or narrow multiple-choice patterns, the score is not a carrier for broad legal reliability. The strongest status is benchmark-legitimate or evidence-compatible, not deployment-reliable. Final status: benchmark-legitimate within the declared benchmark, or evidence-compatible for the broader legal-reliability claim. Next action: contamination audit, task-aligned validation, domain-expert review, and deployment-specific risk test.

G.9 AI/ML guide nonclaims

UEAP does not define intelligence, consciousness, or alignment. It audits whether AI/ML claims have the target, carrier, contamination, domain, and reportability conditions required for the status asserted.

H Medicine, biomarkers, and surrogate endpoints domain instantiation guide

H.1 Domain scope

This domain instantiation guide applies UEAP to diagnostic biomarker claims, surrogate endpoint claims, subgroup treatment claims, risk prediction models, retrospective/prospective separation, mechanistic biomedical claims, and clinical decision-support claims. It does not provide medical advice, determine

individual treatment, or replace regulatory standards. Regulatory status can be audited only when the jurisdictional and regulatory standard is declared.

Medical claims are especially vulnerable to target/proxy mismatch. A biomarker may measure a biological process without measuring the clinical endpoint. A surrogate may track disease activity without supporting treatment benefit. A retrospective model may predict its own fitted data without being clinically actionable. UEAP preserves these distinctions as status differences rather than treating them as generic uncertainty.

H.2 Typical medical claim forms

Table 26: Medical claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Biomarker	Marker X indicates disease Y .	proxy laundering
Surrogate endpoint	Endpoint S predicts clinical outcome O .	surrogate laundering
Treatment effect	Treatment T improves outcome O .	confounding; endpoint drift
Risk score	Score R predicts event E .	calibration-only
Mechanism	Pathway P causes disease.	post-hoc explanation
Actionability	Use X to guide therapy.	report-status overreach
Subgroup claim	Treatment works for subgroup G .	post-hoc subgroup laundering

H.3 Standing-bearing carriers in medicine

Medical carriers include randomized controlled trials, prospective cohorts, validated diagnostic studies, causal models with confounding control, biological mechanisms plus empirical validation, external validation cohorts, regulatory-grade endpoint analysis, and meta-analyses with independence audits. Correlation may support association, not diagnosis or causation. Retrospective subgroup discovery is not prospective treatment guidance. Internal validation is not external validity. Surrogate endpoints do not automatically carry clinical-outcome standing.

H.4 Required medical domain ledger

Table 27: Medical domain ledger.

Ledger field	Required audit detail
Population	Inclusion/exclusion criteria, demographics, disease stage, setting.
Endpoint	Clinical, surrogate, biomarker, composite, or action threshold.
Intervention/exposure	Dose, timing, comparator, adherence, co-interventions.
Study design	RCT, prospective cohort, case-control, retrospective, registry.
Confounders	Measured and unmeasured confounding discipline.

Ledger field	Required audit detail
Validation	Internal, external, prospective, cross-site, or regulatory validation.
Clinical action	Diagnostic, prognostic, therapeutic, triage, or monitoring use.
Report object	Association, risk, diagnosis, causal effect, clinical benefit, or actionability.

H.5 Common medical freedoms

Common freedoms include population freedom, endpoint freedom, surrogate-target freedom, subgroup freedom, confounder freedom, measurement freedom, follow-up-window freedom, treatment-regime freedom, retrospective/prospective freedom, and reportability/actionability freedom. A claim must preserve these ledgers to move from association to diagnosis, prognosis, treatment effect, or clinical actionability.

H.6 Medical laundering routes

Table 28: Medical laundering routes.

Route	Medical manifestation	Blocker
Proxy laundering	Biomarker treated as disease outcome.	proxy blocker
Surrogate laundering	Surrogate endpoint treated as clinical benefit.	calibrated/domain-dependent
Subgroup laundering	Post-hoc subgroup treated as confirmed.	calibrated-only
Endpoint laundering	Composite endpoint hides weak component.	endpoint blocker
Authority laundering	Guideline or expert opinion replaces evidence carrier.	authority-only
Computation laundering	Risk score treated as clinical truth.	computation-only
Domain laundering	Trial population generalized to excluded patients.	domain-dependent
Mechanism laundering	Plausible pathway treated as outcome proof.	evidence-compatible only
Actionability laundering	Association reported as treatment guidance.	not actionable

H.7 Medical status assignment rules

A legitimate clinical claim requires appropriate endpoint, population, study design, validation, and report object. A calibrated-only claim is internally fitted or target-derived without external or prospective validation. A domain-dependent claim is limited to trial population, measurement context, or treatment regime. An evidence-compatible claim has biological plausibility or observational association but not the requested clinical status. A fixed association may be legitimate but not actionable if no validated decision threshold or treatment linkage is established.

H.8 Miniature worked audit

Claim: “Biomarker X can be used to guide treatment T .” The target is treatment-guidance actionability. The carrier is a biomarker study. Active freedoms include population, clinical endpoint, surrogate relation, treatment interaction, measurement protocol, and validation status. If X is merely correlated with disease, the status is evidence-compatible. If it is an internally fitted risk model, the status is calibrated-only. If prospective validation shows that using X changes clinical outcome under a declared protocol, actionability may become reportable. Final status before such validation: evidence-compatible or calibrated-only, depending on study design. Next action: prospective validation with clinical endpoint and treatment-decision threshold.

H.9 Medical guide nonclaims

UEAP is not a clinical decision system or regulatory substitute. It classifies claim status and identifies missing evidence carriers.

I Law, forensics, and evidence-chain domain instantiation guide

I.1 Domain scope

This domain instantiation guide applies UEAP to forensic evidence claims, witness convergence, reconstruction claims, causation claims, chain-of-custody claims, expert testimony, and probability or risk claims in legal settings. It does not provide legal advice, determine guilt or liability, or replace jurisdiction-specific doctrine. Jurisdiction and legal standard must be part of the domain ledger when legal admissibility or probative status is the target.

Legal and forensic claims often fail by authority laundering, narrative laundering, evidence-network laundering, or source/proxy mismatch. UEAP distinguishes consistency with a source, likelihood under a population, admissibility under a legal standard, and proof of causal or source identity.

I.2 Typical legal and forensic claim forms

Table 29: Legal/forensic claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Identification	Trace X came from source Y .	source/proxy laundering
Causation	Act A caused harm H .	causal gap
Reconstruction	Event occurred as described.	narrative laundering
Chain of custody	Evidence was preserved.	carrier break
Expert claim	Expert concludes C .	authority laundering
Witness convergence	Multiple witnesses agree.	evidence-network laundering
Probability claim	Risk or likelihood is p .	baseline and population freedom

I.3 Standing-bearing carriers in law and forensics

Standing-bearing carriers include chain of custody, validated forensic method, expert method disclosure, witness testimony with credibility and independence assessment, physical evidence with provenance, causal reconstruction, admissibility standard, and probability model with declared assumptions. Expert authority is not method standing. Multiple witnesses are not independent if contaminated by common source. A forensic match is not source certainty unless error rates, comparison class, and population basis are fixed.

I.4 Required legal/forensic domain ledger

Table 30: Legal/forensic domain ledger.

Ledger field	Required audit detail
Jurisdiction/standard	Admissibility rule, burden standard, permitted inference type.
Evidence item	Identity, provenance, preservation, chain of custody.
Method	Validation, error rate, limitations, comparison class.
Source hypothesis	Claimed source, event, actor, or causal relation.
Alternative hypotheses	Admissible rival explanations and exclusion status.
Witness independence	Contact, contamination, common source, memory conditions.
Expert carrier	Method, qualifications, scope, uncertainty, and report limits.
Report object	Admissible, probative, causal, identification, reconstruction, or compatible.

I.5 Common legal/forensic freedoms

Common freedoms include source identity freedom, chain-of-custody freedom, method-validation freedom, comparison-class freedom, witness-contamination freedom, causal-path freedom, narrative-selection freedom, jurisdiction-standard freedom, probability-interpretation freedom, and reportability freedom.

I.6 Legal and forensic laundering routes

Table 31: Legal/forensic laundering routes.

Route	Manifestation	Blocker
Authority laundering	Expert status replaces method validation.	authority-only
Evidence-network laundering	Non-independent witnesses treated as independent.	evidence-compatible only
Proxy laundering	Trace similarity treated as source identity.	proxy blocker
Narrative laundering	Coherent story treated as proof.	compatible-only
Method laundering	Lab procedure treated as validated science.	carrier-unfixed

Route	Manifestation	Blocker
Baseline laundering	Unspecified population frequency or base rate.	metric/baseline blocker
Domain laundering	Method used outside validated domain.	domain-dependent
Post-hoc explanation	Reconstruction fitted to known outcome.	calibrated-only

I.7 Legal/forensic status assignment rules

A legitimate evidentiary claim requires provenance, method, alternatives, legal standard, and report object to be fixed. Evidence-compatible status is appropriate when evidence supports a claim without eliminating alternatives. Domain-dependent status applies when a method is valid only under declared conditions. Authority-only or carrier-unfixed applies where expert conclusion lacks method carrier. Narrative compatibility is not source identity, causation, or proof.

I.8 Miniature worked audit

Claim: “Forensic trace X identifies suspect Y .” The target is source identification. The carrier is the forensic method plus chain of custody. Active freedoms include comparison population, error rate, contamination, alternative sources, and reporting standard. If the method establishes similarity only, the status is evidence-compatible. If it supplies a validated likelihood ratio under a declared population and chain, the report may be stronger but remains tied to that population and method. If the chain of custody is broken, the carrier is unfixed or failed. Final status: evidence-compatible unless the method, population, error rate, and chain of custody support a stronger domain-specific evidentiary status. Reportable form: “Trace X is consistent with Y under method M , with limitations L .” Next action: declare the comparison population, error model, chain status, and legally relevant reporting standard.

I.9 Legal/forensic guide nonclaims

UEAP does not determine guilt, liability, or jurisdiction-specific admissibility. It audits standing and reportability of evidentiary claims.

J Engineering, safety, and simulation domain instantiation guide

J.1 Domain scope

This domain instantiation guide applies UEAP to engineering safety margins, simulation validity, load and failure claims, reliability claims, boundary-condition claims, certification-readiness claims, degradation claims, digital-twin reports, and failure-mode coverage. It does not replace engineering judgment, certification bodies, or standards. Regulatory or certification claims must declare the standard as part of the domain ledger.

Engineering claims frequently fail through computation laundering, boundary laundering, hidden failure modes, scale-transfer gaps, or operating-envelope drift. UEAP preserves the difference between “computed”, “tested”, “validated”, “safe within envelope”, and “certified under standard”.

J.2 Typical engineering claim forms

Table 32: Engineering claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Safety margin	Design withstands load L .	boundary/domain laundering
Simulation validity	Simulation proves safety.	computation laundering
Reliability	System failure rate is p .	data/model mismatch
Certification readiness	Design meets standard S .	standard drift
Failure coverage	All critical modes are covered.	hidden remainder
Scaling	Lab result transfers to full scale.	transport freedom
Durability	System remains safe over time T .	degradation ledger gap

J.3 Standing-bearing carriers in engineering

Engineering carriers include validated simulation, physical test, certified calculation, material specification, operating envelope, failure-mode analysis, safety-factor standard, inspection/quality ledger, and digital twin with validation evidence. Simulation without validation is computation-only. A safety margin is domain-dependent unless the operating envelope is fixed. A failure-mode coverage claim requires a residual-risk and hidden-remainder ledger. Scaling requires a transport certificate or declared scope restriction.

J.4 Required engineering domain ledger

Table 33: Engineering domain ledger.

Ledger field	Required audit detail
System boundary	Component, subsystem, system, environment, interfaces.
Load envelope	Magnitudes, cycles, extremes, combinations, uncertainty.
Materials	Properties, tolerances, fatigue, degradation, manufacturing variation.
Model assumptions	Geometry, boundary conditions, solver, mesh, time step, constitutive laws.
Validation data	Test regime, calibration, uncertainty, cross-validation.
Failure modes	Enumerated modes, coverage, residuals, hidden modes.
Safety factor	Standard, basis, applicability, margin interpretation.
Report object	Safe, compliant, robust, simulated, tested, certified, or domain-dependent.

J.5 Common engineering freedoms

Common freedoms include load-envelope freedom, boundary-condition freedom, material-property freedom, solver/mesh freedom, scale-transfer freedom, degradation/time freedom, failure-mode coverage freedom, safety-factor freedom, operating-regime freedom, and reportability freedom. Each freedom must be declared, closed, or retained as a blocker.

J.6 Engineering laundering routes

Table 34: Engineering laundering routes.

Route	Manifestation	Blocker
Computation laundering	Unvalidated simulation treated as proof.	computation-only
Boundary laundering	Favorable boundary condition hidden.	domain-dependent
Endpoint laundering	Test endpoint selected to pass.	endpoint blocker
Scale laundering	Model-scale result treated as full-scale.	transport blocker
Metric laundering	Safety factor chosen post hoc.	baseline blocker
Evidence-network laundering	Redundant tests treated as independent.	evidence-compatible only
Authority laundering	Certification label without scope.	authority-only
Hidden remainder	Unmodeled failure mode ignored.	failed/obstruction-only

J.7 Engineering status assignment rules

A legitimate safety claim requires fixed operating envelope, adequate carrier, validation, failure-mode coverage, and report-preserving wording. Domain-dependent status applies when safety holds only under specified envelope. Computation-only applies when a simulation lacks validation or boundary closure. Evidence-compatible applies when tests support but do not cover the failure space. Obstruction-only applies when a failure route is ruled out without a positive safety claim. Failed applies when hidden failure modes or boundary mismatch remain load-bearing.

J.8 Miniature worked audit

Claim: “The structure is safe under expected operating conditions.” The target is safety under envelope E . The carrier is simulation plus test data plus safety standard. Active freedoms include load envelope, material degradation, boundary conditions, solver/mesh choice, and failure modes. If only simulated, the status is computation-only. If tested inside a narrow envelope but extremes remain untested, the status is domain-dependent. If all critical modes are covered under declared standard S , the reportable form is “safe under envelope E and standard S , not outside them.” Final status: computation-only, domain-dependent, or legitimate-within-envelope depending on validation and failure-mode coverage. Next action: close the load envelope, material degradation, solver validation, and critical-failure-mode ledgers.

J.9 Engineering guide nonclaims

UEAP does not replace engineering judgment, standards, certification bodies, or safety culture. It audits whether the safety claim has the status it is being reported to have.

K Forecasting and economics domain instantiation guide

K.1 Domain scope

This domain instantiation guide applies UEAP to forecasts, backtests, economic model claims, causal variable claims, policy-effect claims, market prediction claims, risk-model claims, and scenario claims. It excludes investment advice, policy endorsement, and claims depending on undeclared normative goals. A policy recommendation or investment action is reportable only if the actionability target and domain standard are declared.

Forecasting claims often fail through target-window drift, information-set contamination, backtest selection, metric choice, regime transfer, and scenario/forecast conflation. UEAP separates predictive standing from retrospective fit.

K.2 Typical forecasting and economics claim forms

Table 35: Forecasting/economics claim forms and UEAP risks.

Claim form	Example	Main UEAP risk
Forecast	Variable Y will be v .	calibration/backtest laundering
Causal claim	Policy P causes outcome O .	confounding/proxy
Backtest	Strategy worked historically.	selection laundering
Risk model	Risk is p .	regime/domain drift
Scenario	If A , then B .	boundary dependence
Economic mechanism	Factor F drives trend T .	post-hoc explanation
Policy effect	Intervention I will produce outcome O .	causal-carrier inadequacy

K.3 Standing-bearing carriers in forecasting and economics

Carriers include pre-registered forecasts, out-of-sample validation, causal identification designs, structural economic models, calibrated models with declared status, backtests with leakage controls, scenario analyses with boundary ledgers, and ensembles with independence audits. A backtest supports historical compatibility unless leakage and selection are controlled. A forecast requires time-prior target fixation and information-set discipline. A causal claim requires an identification carrier, not mere correlation. A scenario result is domain-dependent by design.

K.4 Required forecasting/economics domain ledger

Table 36: Forecasting/economics domain ledger.

Ledger field	Required audit detail
Forecast target	Variable, horizon, event definition, unit, threshold.
Information set	Data available at forecast time, leakage exclusions.
Model class	Statistical, structural, causal, agent-based, or hybrid.
Regime	Market, economic, policy, institutional, or macro regime.
Backtest protocol	Window, selection, rebalancing, survivorship, look-ahead exclusion.
Error metric	Loss, scoring rule, calibration measure, confidence interpretation.
Causal identification	Instruments, controls, natural experiment, assumptions, robustness.
Report object	Forecast, scenario, causal estimate, risk estimate, compatibility result.

K.5 Common forecasting/economics freedoms

Common freedoms include target-window freedom, information-set freedom, backtest-selection freedom, survivorship freedom, causal-identification freedom, metric/loss freedom, regime-transfer freedom, scenario-boundary freedom, proxy-variable freedom, and reportability freedom. These freedoms are especially dangerous because they often remain invisible after a forecast is summarized by one number.

K.6 Forecasting and economics laundering routes

Table 37: Forecasting/economics laundering routes.

Route	Manifestation	Blocker
Backtest laundering	Selected history treated as prediction.	calibrated-only
Proxy laundering	Correlated indicator treated as cause.	proxy blocker
Regime laundering	Old regime generalized to new regime.	domain-dependent
Metric laundering	Favorable loss function selected.	baseline blocker
Target-drift laundering	Forecast target changed after outcome.	failed/blocked
Authority laundering	Institutional forecast treated as proof.	authority-only
Scenario laundering	Conditional scenario reported as forecast.	reportability failure
Evidence-network laundering	Correlated indicators treated as independent.	evidence-compatible only

K.7 Forecasting/economics status assignment rules

A legitimate forecast requires target fixation before outcome, declared information set, fixed scoring rule, and out-of-sample or prospective status appropriate to the claim. A calibrated-only claim is fitted or backtested after target knowledge. A domain-dependent claim is valid only under declared regime or scenario. Evidence-compatible status applies when indicators agree but do not eliminate alternatives.

Relation-only status applies when a relation is fixed but a point forecast is not. Failed applies when target drift or leakage invalidates the claim.

K.8 Miniature worked audit

Claim: “This model predicts recessions twelve months ahead.” The target is recession occurrence at a twelve-month horizon. The carrier is the model plus pre-registered forecast record or out-of-sample validation. Active freedoms include recession definition, target horizon, information set, backtest window, threshold, scoring rule, and regime transfer. If the model was selected after historical backtesting, the status is calibrated-only. If a prospective out-of-sample record exists under fixed scoring, the claim may be a legitimate forecast within scope. If it is conditional on an assumed policy or macro regime, the status is domain-dependent. Final status: calibrated-only for selected backtests, domain-dependent for scenario results, or legitimate forecast within scope for prospective out-of-sample records under fixed scoring. Next action: freeze the target, information set, scoring rule, and evaluation horizon before outcome observation.

K.9 Forecasting/economics guide nonclaims

UEAP does not make forecasts, choose policies, or provide investment advice. It classifies whether forecasting and economic claims have standing, calibration independence, domain stability, and reportable status.

L Cross-domain failure crosswalk

The domain instantiation guides instantiate the same UEAP blockers across different carrier standards. Table 38 records the recurrent pattern. The table is not an additional theorem; it is an operational map showing how the coordinate-exhaustion theorem appears in practice. The crosswalk is not a reduction of domain standards to one vocabulary; it shows that different domain failures instantiate recurring audit roles.

Table 38: Cross-domain failure crosswalk.

UEAP blocker	Domain manifestations
Target-unfixed	Science: vague phenomenon. AI/ML: vague capability. Medicine: unclear endpoint. Law/forensics: unclear source or causation claim. Engineering: unclear safety envelope. Forecasting: vague target horizon or event definition.
Carrier-unfixed	Science: unsupported model. AI/ML: benchmark used for wrong capability. Medicine: weak study design. Law/forensics: invalid or undeclared method. Engineering: unvalidated simulation. Forecasting: backtest without information-set discipline.
Calibrated-only	Science: fitted model. AI/ML: training leakage or prompt tuning. Medicine: retrospective subgroup. Law/forensics: post-hoc reconstruction. Engineering: calibrated simulation without validation. Forecasting: selected backtest.

UEAP blocker	Domain manifestations
Domain-dependent	Science: limited regime. AI/ML: narrow benchmark or threat model. Medicine: trial population. Law/forensics: jurisdiction or method scope. Engineering: operating envelope. Forecasting: economic regime or scenario.
Evidence-compatible	Science: consilience without alternative closure. AI/ML: demo suite. Medicine: association. Law/forensics: witness consistency. Engineering: partial tests. Forecasting: correlated indicators.
Computation-only	Science: solver output. AI/ML: generated benchmark score without audit. Medicine: risk score. Law/forensics: forensic software output. Engineering: simulation. Forecasting: model output.
Laundered/blocked	Science: post-hoc theory. AI/ML: benchmark or prompt laundering. Medicine: surrogate endpoint laundering. Law/forensics: authority or narrative laundering. Engineering: boundary laundering. Forecasting: backtest or regime laundering.
Reportability failure	Science: compatibility reported as confirmation. AI/ML: benchmark score reported as broad capability. Medicine: association reported as actionability. Law/forensics: consistency reported as identification. Engineering: safe-within-envelope reported as universally safe. Forecasting: scenario reported as forecast.

L.1 Use of the crosswalk

The crosswalk should be used in two directions. First, when auditing a domain-specific claim, it suggests likely blockers. Second, when comparing claims across domains, it prevents false equivalence. A computation-only engineering claim and a computation-only AI benchmark claim are not the same evidence carrier, but they share the same status logic: computation has occurred, while the carrier and reportability conditions for the stronger claim remain unresolved.

L.2 Domain guides as deployment layers

The domain instantiation guides do not introduce new foundational conditions. They instantiate UEAP’s general coordinates: target, carrier, semantic content, representation, ancestry, domain, evidence network, report/use profile, requested status, blocker set, and next action. A future domain instantiation guide may refine carrier standards, but it must still report how its refinements map onto these coordinates.

M Anticipated objections and scope clarifications

Table 39: Anticipated objections and scope clarifications.

Objection	UEAP response
This is just a checklist.	The checklist operationalizes target fixation, carrier standing, freedom exhaustion, laundering exclusion, and report preservation. The dependency order is fixed by the declared audit basis; changing it changes the protocol, not merely its presentation.
The paper smuggles in AASC.	The protocol is operationally self-contained. The kernel bridge supplies rationale, but the validator and status taxonomy can be applied independently.
Truth is separate from legitimacy.	UEAP audits claim standing and reportability, not truth itself.
Useful claims may fail UEAP.	They receive compatible-only, calibrated-only, domain-dependent, computation-only, or other weaker statuses.
Different domains require different evidence standards.	UEAP fixes the status logic, not the domain-specific carrier content.
The biconditional is too strong.	The biconditional is internal to UEAP-legitimacy under a declared audit basis. σ fixes the requested report class and its domain evidence standard; UEAP does not erase domain expertise.
Theorems look definitional.	The manuscript separates definitions, conditional protocol theorems, and kernel theorems. The protocol theorems state what follows once the audit basis, coordinates, and status-precedence rules are fixed.
The pilots are too different.	Their heterogeneity is intentional: the same target/carrier/freedom/laundrying/reportability roles recur across valuation, operator extension, and empirical inference.
A computed result should count as evidence.	It can count as evidence, but not automatically as standing. The computation’s target, carrier, inputs, alternatives, and reportability still require audit.
Authority should matter.	Authority can be part of evidence or carrier in domains where authority is itself the target. It cannot substitute for claim standing when the target is factual, mathematical, causal, or predictive.

N Declarations and source status

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing interests. The author declares no competing interests.

Data availability and source status. No datasets were generated or analyzed for the theoretical protocol. Pilot examples are drawn from separately cited source documents and are used only as classification examples. The cosmological pilot is summarized only within the conditional limits stated in the cited transport-closure source document. The protocol-level claims of the present paper do not depend on the pilots as independent validation. The cited source documents identify the analyses from which the worked classifications are drawn; they are not used as independent authority for the protocol-level theorems. The present manuscript is the UEAP paper itself; the exactness arc remains an unstabilized source document until an archival record is created. Where stable archival records are available for other source documents, Zenodo DOI and record information is supplied in the bibliography.

AI/editorial assistance note. AI-assisted tools were used for editorial hardening, organization, readability, LaTeX assembly, and language refinement of human-directed manuscript material. No AI-assisted tool is listed as an author. The author reviewed and is responsible for the final manuscript content.

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